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# Recycled cigarette butts as a sustainable additive in PAN-based nanofiber membranes for high-efficiency air filtration

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ABSTRACT

Cigarette butts (CBs), the world's most abundant plastic litter, pose severe environmental threats due to their non-biodegradable cellulose acetate (CA) content. This study presents a circular economy solution by recovering CA from discarded CBs and co-electrospinning it with polyacrylonitrile (PAN) to fabricate high-performance air filtration membranes. Systematic optimization of PAN:CBs ratios and total solid content (8-18 wt%) revealed that hybrid nanofibers with 10 wt% PAN and 6 wt% CBs achieved exceptional $PM_{1.0}$ filtration efficiency (99.99%), low pressure drop (173 Pa), and a high quality factor (0.0558 $Pa^{-1}$), outperforming pure PAN counterparts. The incorporation of waste-derived CA not only enhanced specific surface area and fiber morphology but also reduced raw material costs by 27%. Coupled with a techno-economic assessment, this work demonstrates a scalable, eco-efficient pathway to transform hazardous urban waste into value-added functional materials, aligning with circular economy principles for simultaneous mitigation of plastic pollution and air quality degradation.

## 1. Introduction

Cigarette butts (CBs) represent the most discarded form of plastic litter globally, with an estimated 4.5 trillion discarded annually out of the 6 trillion cigarettes consumed worldwide (Torkashvand and Farzadkia, 2019). According to methodologically robust organizational audits, such as the Keep America Beautiful (2020) survey (Tobacco and the environment Apr. 26, 2023), estimates that this staggering volume translates to approximately 9.7 billion tons of waste, posing serious ecological and public health risks when improperly disposed of. Most CBs are made of cellulose acetate (CA), a type of non-biodegradable plastic that can persist in the environment for decades (Kurmus and Mohajerani, 2020). During this time, they leach harmful chemical substances, including nicotine, heavy metals (such as cadmium and lead), and polycyclic aromatic hydrocarbons (PAHs), into soil and aquatic ecosystems (Torkashvand and Farzadkia, 2019). As global cigarette consumption and cigarette waste continues to rise, effective management strategies remain challenging (De Fenzo et al., 2020). Therefore, CBs offer a compelling opportunity for circular economy initiatives,

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given their potential for valorization through chemical recycling and repurposing into high-value materials (Kurmus and Mohajerani, 2021).

The need for effective particulate matter (PM) filtration has grown increasingly urgent as air pollution levels rise, especially in urban areas (Flood-Garibay et al., 2023). PM, including $PM_{10}$, $PM_{2.5}$, and $PM_{1.0}$, consist of microscopic airborne particles that pose serious health risks, such as lung cancer (Giammona et al., 2024), chronic respiratory conditions, and heart disease, leading to increased healthcare burdens and reduced life expectancy (Nakhjirgan et al., 2023). The World Health Organization (WHO) identifies PM as one of the most significant environmental health threats worldwide, with the heaviest burden falling on densely populated cities ((WHO) 2021). To protect individuals from the harmful effects of PM, two approaches have been proposed: reducing PM emissions at the source and employing filtration methods. Comparatively, the latter method proves superior, offering a swift and effective means of shielding people from PM pollutants. In light of these severe public health consequences, there is an urgent need for innovative, high-performance, and sustainable air filtration technologies capable of efficiently capturing both fine and ultrafine particles. Such solutions are vital not only for safeguarding human health but also for addressing the wider environmental and economic costs associated with air pollution.

Aligned with the principle of transforming waste into environmental solutions, CBs have been repurposed across a wide range of application (Akid et al., 2025). Previous literature highlights their use in construction, where they serve as fillers or binders in bricks (Paulista et al., 2024), road paving (Guo et al., 2024), and insulation materials (Maderuelo-Sanz, 2021). In environmental engineering, CBs have been transformed into functional materials for removing pollutants from water, including organic contaminants (Premchand et al., 2025), oils (Doyan et al., 2021), dyes (Fahanwi et al., 2024), and heavy metals (Zhang et al., 2021). Much of these work focuses on converting CBs fibers into activated carbon via methods such as alkaline activation with microwave heating or low-temperature hydrothermal carbonization for applications like methylene blue adsorption (Dalmaz and SİVRİKaya ÖZak, 2024). Despite these promising developments, the utilization of CBs-derived materials for air filtration remains notably underexplored. While electrospun nanofibers are well established for high-efficiency particulate air (HEPA) filtration, most studies focus on biopolymers such as lignin, chitosan, poly (lactic acid) (Russo et al., 2022), polyvinvlyl alcohol (PVA) (Zhang et al., 2019), polyvinylidene fluoride (PVDF) (Zhang et al., 2022) and polyacrylonitrile (PAN) (Lakshmanan et al., 2021). To date, only Kakoria et al. have demonstrated air filtration using cellulose acetate recovered from waste cigarette butts (Kakoria and Sinha-Ray, 2022). By employing supersonic solution blowing, they fabricated ultrafine nanofibers with a mean diameter of approximately 92.4 nm, achieving a filtration efficiency exceeding 99.7% against $PM_{0.1}$. However, the rapid evaporation of acetone, a highly volatile and flammable solvent prone to forming explosive air mixtures, induced sporadic bead formation during processing. While the resulting membrane exhibited exceptional filtration performance, these morphological defects may compromise fiber uniformity, mechanical robustness, and long-term operational stability, particularly during large scale manufacturing. In contrast, recent studies have successfully leveraged electrospinning to fabricate nanofibrous membranes from alternative feedstocks derived from waste. For instance, Özen et al., (2025) developed hemp hurd and polyvinyl alcohol composite membranes. Their optimal formulation with 15 wt% hemp hurd achieved filtration efficiencies of 98.74% for $PM_{2.5}$ and 98.73% for $PM_{10}$, alongside a quality factor of 0.049 inverse pascals. Similarly, Scaffaro et al., (2026) engineered a sandwich structured waste wool and polylactic acid membrane that delivered superior filtration efficiency with minimal pressure drop, yielding a higher quality factor of 0.079 inverse pascals. Collectively, these findings demonstrate that integrating materials derived from waste with biopolymers, particularly *via* electrospinning, can simultaneously enhance filtration efficiency, reduce pressure drop, and optimize the overall quality factor, offering a sustainable and scalable pathway for next generation air filters. Crucially, no study has yet explored the use of CBs-derived CA in conventional electrospun nanofiber membranes for air filtration, a critical gap, given that electrospinning offers superior control over fiber morphology, diameter, and alignment, enabling precise tuning of pore structure and surface functionality. The blending of CBs-derived CA with polyacrylonitrile (PAN) has the potential to create hybrid nanofiber membranes with enhanced air filtration properties. PAN, a widely used polymer in electrospinning, is prized for its excellent spinnability, mechanical robustness, and thermal stability, together with good efficiency for gaseous pollutants filtration in respect to other polymer fibers (Zhang et al., 2023a). However, its reliance on petroleum-based sources and high production cost Kakorias remain limitations. Integrating waste-derived CA into PAN could reduce the demand for virgin polymer materials, while introducing polar ester and hydroxyl groups that may enhance surface-particle interactions, thereby improving filtration efficiency. Recent studies have suggested that the hybridization of natural fibers with synthetic polymers can lead to synergistic effects in nanofiber properties, including improved mechanical performance and filtration efficiency (Deng et al., 2021; Wang et al., 2016). Moreover, despite growing interest in waste-to-resource strategies, there remains a lack of systematic techno-economic assessments of CBs-derived air filtration materials. Critical performance indicators, such as filtration efficiency, pressure drop, quality factor, production cost, and dust-holding capacity, are essential for evaluating both functional efficacy and commercial viability. These metrics not only reflect a filter's ability to capture particulate matter under realistic operating conditions but also determine its service life, energy consumption, and scalability. To date, such comprehensive evaluations are largely absent in studies on CBs-based filters, highlighting an urgent need for integrated performance and cost analyses to bridge the gap between laboratory innovation and real-world deployment.

To address these gaps, this study presents the comprehensive investigation into the fabrication of hybrid PAN/CBs nanofiber membranes for air filtration applications. Given the potential toxicity of waste CBs, the cleaned samples were subjected to comprehensive characterization using FT-IR, GPC, XPS, ICP-MS, and GC-MS. These analyses confirmed the effective removal of contaminants and substantiated the environmental safety of the treated material. PAN serves as the structural matrix, and a series of PAN/CBs nanofiber membranes were fabricated via blend electrospinning under systematically varied formulation conditions: series A: fixed total concentrations of 8 wt% with varying PAN:CBs mass ratios 1:0 to 1:1, series B: fixed total concentrations of 10 wt%, and 14 wt% with varying PAN:CBs mass ratios 1:0 and 1:1, series C: PAN-only solutions at concentrations 8-18 wt% in 2 wt% increments, no CBs, and series D: fixed PAN mass 8, 10, 14 wt% with increasing CBs to push total solids to 18 wt%. Within the described series of

preparation, a simple, fast, and low-cost procedure was introduced to recover CA from CBs, and to prepare a sustainable membrane sorbent for air filtration purposes. In addition, a selected nanofiber membrane (prepared from the PAN/CBs 8 wt% solution with a 1:1 PAN:CB mass ratio) was further subjected to heat-press post-treatment at 100°C, 120°C, and 140°C to evaluate the impact of thermal densification on fiber morphology and filtration performance. All membranes were thoroughly characterized using SEM-EDS, BET, water contact angle measurement, ATR-FTIR, thickness profiling, tensile strength testing, BET specific surface area, and zeta potential analysis. The air filtration performance is evaluated against incense-generated PM across multiple size fractions, with a focus on achieving high filtration efficiency (>99%) and low pressure drop. Additionally, this work also includes a comprehensive techno-economic analysis, integrating key performance indicators, including filtration efficiency, pressure drop, quality factor, estimated production cost, and dust-holding capacity tests, to evaluate both functional efficacy and commercial viability. By demonstrating that waste CBs can serve not merely as fillers but as performance-enhancing additives in PAN-based nanofiber membranes, this work contributes to both the advancement of sustainable urban waste valorization and could lead to the development of next-generation, eco-efficient air purification technologies.

## 2. Materials and methods

### 2.1. Chemical reagents

Cigarette butts were collected from cigarette butts' collectors located in restaurants, walking road, and buildings from the campus of Shandong University of Science and Technology. Polyacrylonitrile (PAN, $M_w$ = 150,000 g/mol), Cellulose acetate (CA, $M_w$ = 30,000 g/mol) were purchased from Shanghai Macklin Biochemical Technology Co., Ltd, Shanghai, China, whereas N, N-dimethylformamide (DMF, 99.5%) was purchased from Sinopharm Chemical Reagent Co. Ltd. (China), anhydrous ethanol (EtOH, 99.7% w/w) was purchased from Tianjin Fuyu Fine Chemical Co., Ltd, Tianjin, China., and deionized water was self-made by the laboratory. All the materials and reagents were used as received without any purification.

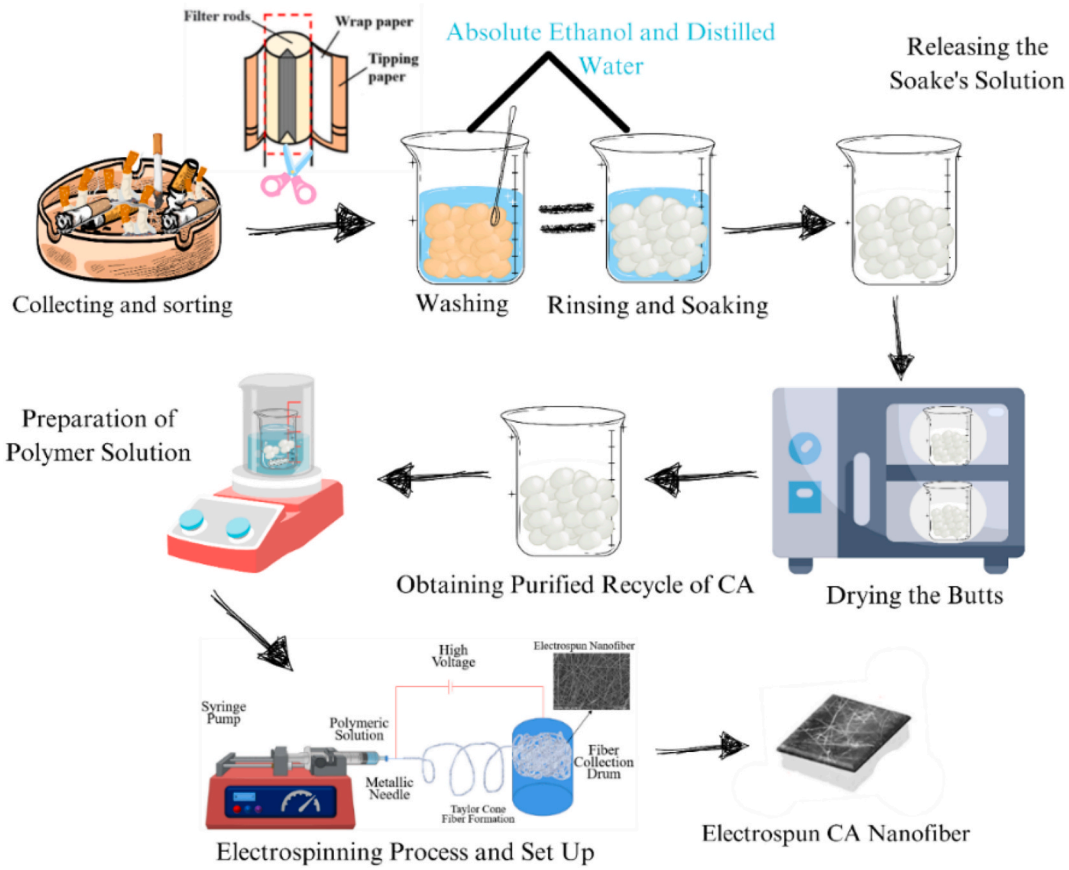



**Fig. 1.** Schematic diagram of preparation process for nanofiber membranes by using recycled CBs.

### 2.2. Extraction and characterization of CA from waste CBs

The procedure depicted in Fig. 1 outlines the method for extracting CA from discarded CBs. The discarded CBs were manually stripped to retrieve the cotton cores. These cores were then carefully cut into coarse fiber segments measuring 1-2 cm, which contained contaminants including nicotine, heavy metals, and priority polycyclic aromatic hydrocarbons (PAHs), thereby requiring further treatment. To obtain pristine regenerated CA, the extracted material underwent a series of pressing, soaking, and washing cycles using deionized water (at a ratio of 1 g CBs to 10 mL water) followed by anhydrous ethanol (at a ratio of 1 g CBs to 20 mL anhydrous ethanol) for a designated period, repeated at least 5-6 times until all traces of dirt were eradicated to solubilize and extract hydrophobic contaminants such as nicotine. Ethanol was chosen for its selectivity toward organic impurities while preserving CA polymer integrity. Once the remnants were thoroughly cleansed, a final rinse with distilled water was administered to eliminate any residual ethanol and water-soluble toxins that may prevent re-adsorption of contaminants during drying. The soaked fibers, or remnants, were then subjected to a drying process in a vacuum dryer set at 65°C for approximately 7-8 h, eliminating any remaining moisture and evaporating the lingering solvent, thereby yielding pure regenerated cellulose acetate segments. Higher temperatures might induce deacetylation or polymer chain scission. It is of utmost importance to consistently cover the recycled remnants when not in use to prevent evaporation of the cellulose acetate. All work should be conducted under a fume hood, and basic protective gear should be worn, including gloves, a mask, and safety goggles, as even small amounts of residual nicotine can be harmful.

Once purified, recycled CA must pass safety checks to ensure no harmful residues remain. To verify this requirement, X-ray photoelectron spectroscopy (Thermo Fisher ESCALAB Xi+) was used to analyze the elemental composition and chemical structure of the CBs. Inductively coupled plasma mass spectrometry (Agilent 7800 ICP-MS, USA) was performed to determine the concentrations of heavy metals, specifically arsenic (As), chromium (Cr), lead (Pb), and cadmium (Cd), in the CBs' samples. Gas Chromatography-Mass Spectrometry (ISQ7000+TRACE1300) was employed to quantify the concentrations of 16 PAHs in CBs before and after the cleaning process. The molecular weight distribution of CA obtained from discarded CBs was measured *via* Gel Permeation Chromatography (GPC) on a PL-GPC120 (Agilent Technologies, USA) instrument equipped with a refractive index (RI) detector.

### 2.3. Membrane fabrication

PAN and CBs polymer solutions were prepared under four different series (series A: fixed total concentrations of 8 wt% with varying PAN:CBs mass ratios 1:0 to 1:1, series B: fixed total concentrations 10, and 14 wt% with varying PAN:CBs mass ratios 1:0 and 1:1, series C: PAN concentrations 8-18 wt% in 2 wt% increments, no CBs and series D: fixed PAN mass 8, 10, 14 wt% with increasing CBs to push total solids to 18 wt%) by dissolving in DMF at 68°C under constant stirring for at least 7 h. During this time, ensure that the mixture thoroughly blends and dissolves, to form a homogeneous mixture. Allow the liquid to rest for a brief period before extracting it using a syringe for the spinning process. Table S1 summarizes specific electrospinning parameters for PAN/CBs electrospinning, whereas the whole procedure is shown in Fig. 1. A 3 mL spinning solution was electrospun directly onto non-woven paper. The PAN/CBs nanofiber was placed in vacuum dryer for at least 7 h at 60°C to completely evaporate the solvent. In addition, a selected nanofiber membrane (prepared from the PAN/CBs 8 wt% solution with a 1:1 PAN:CBs mass ratio) was subjected to heat-press post-treatment using a paper laminator (M&G stationary Inc. China) at a temperature of 100°C, 120°C, and 140°C, to investigate its effect on fiber morphology and filtration performance.

### 2.4. Membrane characterization

Surface morphology and elemental distribution of the nanofiber membranes were characterized using field-emission scanning electron microscopy (FE-SEM, Regulus 8220, Hitachi, Japan) equipped with energy-dispersive X-ray spectroscopy (EDS, ULTIM-Max, Oxford Instruments, UK). Quantitative analysis of fiber diameter was performed using ImageJ software (National Institutes of Health, USA) by randomly measuring at least 50 individual fibers per image across three representative fields of view, and data are reported as mean ± standard deviation (SD). Pore size distribution was determined by applying consistent thresholding parameters to delineate pore boundaries in SEM images, followed by conversion of pore area to equivalent circular diameter, with results expressed as mean pore diameter ± SD based on analysis of no fewer than 100 pores per sample. Membrane surface wettability was indicated by static water contact angle (WCA) measured on a goniometer (Shanghai, Zhongchen, JC2000D2). The samples were dried sufficiently before the measurement, then a dangling droplet of 1 μL of deionized water was carefully dropped on the sample's surface for at least five times at different position. Three positions on a single sample were randomly chosen for measuring and the mean value was used as the test result. Membrane surface zeta potential was measured on an electrokinetic (SurPASS 3, Anton Paar) with 1 mM KCl solution as background solution, 0.05 mol/L HCl and 0.05 mol/L NaOH were used to adjust the pH from 9.5 to 3. Each sample was tested three times and the average was taken as the measured result. The chemical compositions of samples were characterized by a Fourier transform infrared spectroscopy (FTIR, Nicolet iS10, Thermo Scientific, USA) between 400 and 4000 $cm^{-1}$ to confirm the chemical integrity of PAN and the successful incorporation of cigarette butt-derived components (CA). The mechanical properties were performed on a Floor Model Universal Testing System (5300, Instron Corporation, USA). During the test, samples of 130 × 25 mm size were clamped at both ends with 100 mm of interval and pulled in tension, the tensile speed was 100 mm/min, at least five specimens were tested for each sample. Surface areas were calculated using the Branauer-Emmett-Teller (BET) nitrogen absorption-desorption method. Membrane thickness of each sample were measured using a digital micrometer at multiple points ($n \geq 10$) across the sample to ensure consistency and the average was taken as the measure result. Finally, the porosities ($\varepsilon$) of the electrospun PAN and PAN/CB nanofiber membranes were determined using a gravimetric liquid-displacement method with anhydrous ethanol. Specifically, $W_{wet}$

denotes the weight of the membrane after a 1-h immersion in anhydrous ethanol, and $W_{dry}$ is the weight of the dry membrane after drying in a vacuum oven at 50°C for 24 h. The total porosity was calculated according to Equation (1):

$$\varepsilon = \frac{W_{wet} - W_{dry}}{\rho_{alc} \times A \times z} \times 100\% \quad (1)$$

where $\rho_{alc}$ is the density of anhydrous ethanol, A is the effective area of the membrane, and z is the membrane thickness.

### 2.5. Filtration performance testing

The experimental setup used to evaluate the particulate matter (PM) filtration efficiency and pressure drop is schematically illustrated in Fig. 2. The air filtration performance of the electrospun PAN/CBs nanofiber membranes (cut into 40 × 40 $mm^2$ pieces) was evaluated using a custom-built air filtration testing system. During the experiments, the airflow velocities were maintained at constant values of 3.3, 5.3, and 7.3 cm/s. To simulate severely polluted ambient air, PM aerosols were generated via the controlled combustion of sandalwood incense, which yields a high-concentration aerosol rich in organic, insoluble particles across multiple size fractions (i.e., $PM_{1.0}$, $PM_{2.5}$, and $PM_{10}$). Long-term filtration performance was assessed over an extended duration of 130 min under varied environmental conditions, with temperature and relative humidity (RH) precisely regulated by air conditioning units and a humidifier. To evaluate environmental stability, the PAN/CBs nanofiber membranes were preconditioned at extreme temperatures (−50°C and 85°C) for 2 h, followed by $PM_{1.0}$ filtration testing at 75% RH across all the aforementioned airflow velocities. Furthermore, the reusability of the membranes was investigated through five consecutive filtration-cleaning cycles. After each filtration cycle, the membranes were gently washed with a water/anhydrous ethanol mixture (70:30, v/v) to remove the deposited particulate matter, and then air-dried at room temperature for 4 h prior to the subsequent test. The particle removal efficiency, η, is shown by the following formula:

$$\eta = \left(1 - \frac{C_{down}}{C_{up}}\right) \times 100\% \quad (2)$$

where $C_{up}$ and $C_{down}$ are the mass concentrations of PM before and after the filtration (μg $m^{-3}$).

The quality factor (QF, $Pa^{-1}$) was then determined to balance filtration efficiency and flux, which is calculated using the following formula:

$$QF = \frac{-\ln(1-\eta)}{\Delta P} \quad (3)$$

where η is the filtration efficiency for PM and Δ P is the pressure drop before and after the air filtration (Pa). All of the tests were conducted in triplicate under controlled laboratory conditions (25 ± 2°C, 40–50% RH) to ensure reproducibility (Fan et al., 2025; Sun et al., 2025).

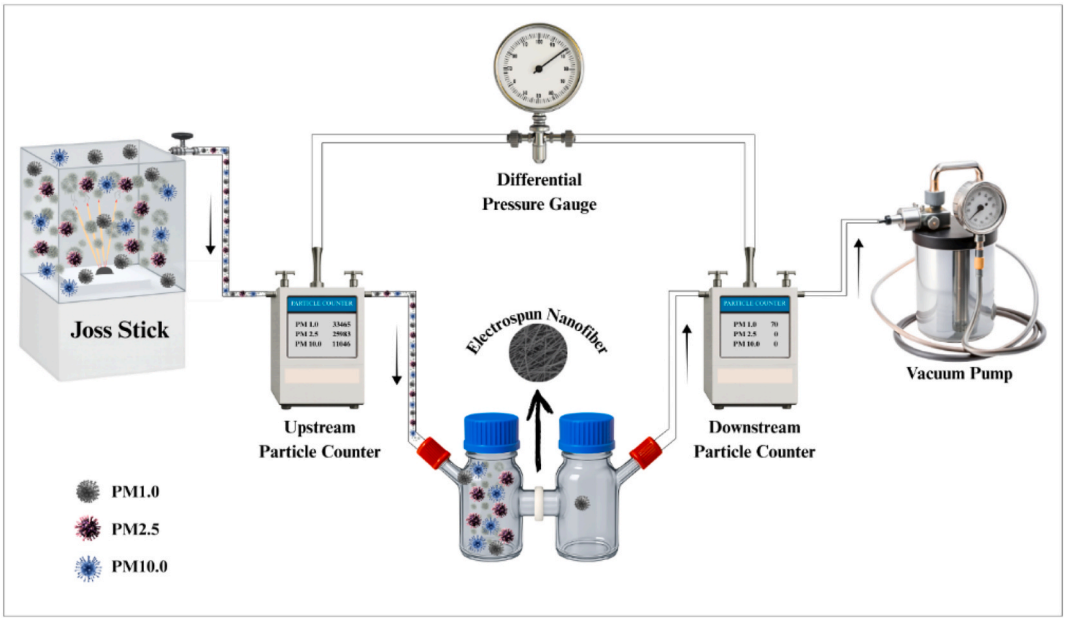



**Fig. 2.** Schematic diagram of the equipment to the evaluate the air filtration performance of nanofiber membranes.

### 2.6. Economic evaluation of PAN/CBs nanofiber membrane

Beyond initial filtration efficiency and pressure drop, the long-term economic viability of air filters is critically determined by their dust-holding capacity and the evolution of flow resistance during operation. To evaluate the practical cost-effectiveness of the developed PAN/CBs nanofiber membranes, dust-loading tests were conducted under controlled aerosol conditions. The maximum dust-holding capacity ($C_{max}$), defined as the dust loading at which the pressure drop reaches a practical upper limit (in this study, $\Delta P_{max} = 450$ Pa), was calculated as (Lee et al., 2025):

$$C_{max} = (m_{max} - m_0) / A \tag{4}$$

where $m_0$ and $m_{max}$ represent the weights of nanofiber membranes without and with dust, and $A$ is the effective area of nanofiber membranes. Accordingly, the max dust capacity ratio, $R_{dust}$, introduced and defined as the ratio of $C_{max}$ for the PAN/CBs membrane to that of the pure PAN membrane:

$$R_{dust} = C_{max,PAN/CBs} / C_{max,PAN} \tag{5}$$

Finally, the cost ratio, $R_{cost}$, is defined to quantify the economic benefit of incorporating recycled CBs. It is calculated as the material cost required to prepare the PAN/CBs nanofiber membrane relative to that of the pure PAN membrane exhibiting the optimal quality factor:

$$R_{cost} = Cost(PAN / CBs) / Cost(PAN) \tag{6}$$

The material cost includes the consumption of both solute, either PAN or CA derived from discarded CBs, and solvent (DMF). The operational cost of the electrospinning process is assumed to be identical across all samples, as they were processed under the same conditions. Material costs are based on industrial market prices: PAN at ¥35-50/kg, DMF at ¥3.5-4.5/kg, and recycled cellulose acetate from CBs at ¥2.0-3.0/kg. Notably, DMF is not recovered in this analysis, as it is fully utilized to prepare the polymer spinning solution and undergoes evaporation during electrospinning without a dedicated recovery infrastructure. In contrast, the anhydrous ethanol used for the initial washing and cleaning of CBs can be efficiently recovered via distillation and reused in the following batches, thereby reducing the effective pretreatment solvent cost. The quoted price range of ¥2.0-3.0/kg for recycled CA is a realistic industrial estimate, as it already covers all steps, from collecting cigarette butts to producing purified cellulose acetate. For the purpose of this

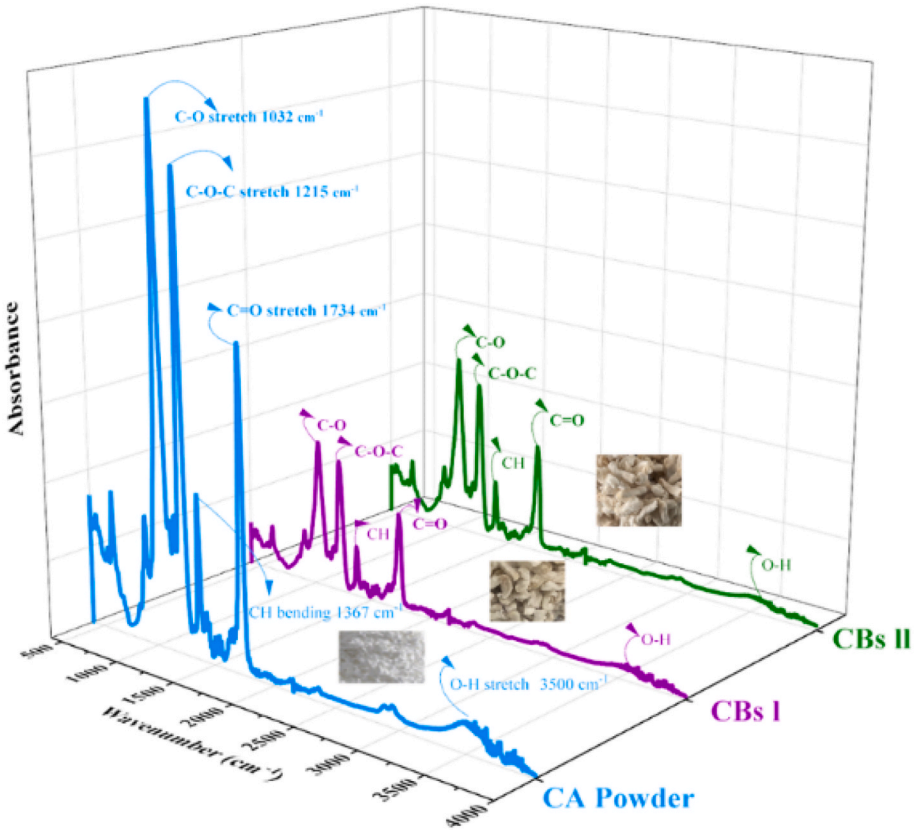



**Fig. 3.** FT-IR spectra of pure CA and recycled CBs. CBs I and CBs II represent cigarette butts collected in two batches.

analysis, representative unit prices of ¥40/kg for PAN, ¥2.5/kg for CBs-derived cellulose acetate, and ¥4.0/kg for DMF are used in next section.

## 3. Results and discussion

### 3.1. Sustainable evaluation of recycled CA from waste CBs

Two independent extraction trials were conducted to recover CA from waste CBs, and the corresponding yields are presented in Fig. S1. Both trials achieved a consistent recovery rate of approximately 79.9% CA, demonstrating high reproducibility under the applied experimental conditions. This consistent yield indicates that the selected solvent system and extraction protocol effectively and selectively dissolve CA while leaving behind non-acetate components such as paper, residual tobacco, and filter additives. The reproducibility across both trials further underscores the robustness of the method, supporting its potential for scale-up in recycling initiatives aimed at valorizing CBs waste.

FT-IR spectra of the recovered CA from both batches were compared with those of pure CA, as shown in Fig. 3. Key characteristic bands of cellulose acetate are clearly evident in the recycled samples. Notably, the strong and sharp carbonyl (C═O) stretching vibration of the acetate group appears at 1734 $cm^{-1}$, an isolated and diagnostic region in the FT-IR spectrum (Yue et al., 2022). Additional characteristic peaks include the asymmetric C–O–C stretching (associated with ester linkages in the glycosidic backbone) at 1215 $cm^{-1}$, the C–O stretching vibration of the ester linkage at 1032 $cm^{-1}$, and C–H bending vibrations at 1367 $cm^{-1}$. A weak O–H stretching band is observed around 3500 $cm^{-1}$, consistent with partial retention of hydroxyl groups; however, its reduced intensity reflects the high degree of acetylation, confirming successful substitution of hydroxyl groups during CA synthesis. Therefore, the spectral similarity between the recycled material and pure CA confirms that the extracted product retains the essential chemical structure of virgin CA, supporting its potential as a viable substitute in future applications (Torkashvand et al., 2021). Furthermore, GPC analysis of CA extracted from waste CBs, as shown in Table S2, was conducted using DMF as the mobile phase and calibrated against narrow polystyrene (PS) standards. The average molecular weights, reported as polystyrene-equivalent values, are: number-average molecular weight ($M_n$) = 52,515 g/mol, weight-average molecular weight ($M_w$) = 180,382 g/mol, and peak molecular weight ($M_p$) = 150, 908 g/mol. The polydispersity index (PDI = Mw/Mn) is 3.43, indicating a broad molecular weight distribution, commonly observed in polymers recovered from post-consumer waste due to partial degradation during use, environmental exposure, or extraction process variability. These molecular weights are sufficient for electrospinning, though the high dispersity may affect solution viscosity and fiber uniformity.

To comprehensively verify safety compliance of the recycled cellulose acetate, surface chemistry, heavy metal content, and organic contaminants were systematically characterized by XPS, ICP-MS, and GC-MS, respectively. XPS analysis was firstly employed to determine surface chemical composition of CBs before and after cleaning. The C 1s spectra of used and cleaned CBs reveal dramatic differences (Figures S2 (a-1) and (b-1)). For used CBs, the deconvoluted C 1s spectrum exhibits two peaks: 284.66 eV assigned to C–C/C–H bonds from adsorbed polycyclic aromatic hydrocarbons (PAHs) and tar deposits, and 285.54 eV attributed to C–N bonds from nicotine and nitrogenous tobacco combustion products. Notably, characteristic cellulose acetate peaks are completely obscured by smoke-derived contaminants, indicating extensive surface coverage by hydrophobic tar residues. In contrast, the C 1s spectrum of cleaned CBs exhibits three well-resolved deconvoluted peaks corresponding to cellulose acetate: 284.65 eV (C–C/C–H bonds in the CA polymer backbone), 286.49 eV (C–O functional groups, including ether linkages and secondary hydroxyls), and 288.87 eV (ester carbonyl of acetate groups). The N 1s spectra provide direct evidence of nitrogenous contaminant removal (Figure S2 (a-2) and (b-2). Used CBs exhibit a strong N 1s signal comprising two components: 399.25 eV (pyridinic/pyrrolic nitrogen from nicotine) and 400.33 eV (amide nitrogen from protein-derived compounds). Following cleaning, the N 1s signal intensity decreases by 90-95%, with only a weak residual peak at 398.72 eV (pyridinic/pyrrolic N). This near-complete elimination demonstrates the cleaning protocol's high efficacy in extracting nicotine and other nitrogen-containing contaminants, confirming the material's suitability for nanofiber membrane fabrication with surface chemistry comparable to virgin cellulose acetate. This finding is further corroborated by the GC-MS quantification of nicotine and 16 priority PAHs in CBs before and after cleaning (Table S3). The results demonstrate the highly effective removal of nicotine and low-to-medium molecular weight PAHs, with removal efficiencies exceeding 97% for most detectable compounds, many of which fell below the limit of detection (LOD = 0.20 μg/g) post-cleaning. Notably, although the removal efficiencies for nicotine and naphthalene reached 99.64% and 80.80%, respectively, their residual concentrations remained at 17.55 μg/g and 0.405 μg/g. Meanwhile, benzo [g,h,i]perylene exhibited an anomalous detection pattern, being quantified at 0.426 μg/g post-cleaning despite being undetectable in the pristine samples. Based on these results, core toxic substances such as benzo [a]pyrene were undetected (<LOD), fully complying with the stringent limit of ≤1 mg/kg (1 μg/g) for plastic components in consumer articles stipulated by the EU REACH Regulation (Annex XVII, Entry 50) ((ECHA) 2025). Regarding the residual nicotine (17.55 μg/g), although there is currently no specific mandatory standard targeting static nicotine residues in air filter media, this level is considered toxicologically safe. This conclusion is supported by the United Nations GHS classification cut-off values for mixtures, as the concentration is far below the 1% (10,000 μg/g) trigger threshold (Nations, 2021), and aligns with the exemption principles for trace impurities under the REACH Regulation, remaining well below the 0.1% threshold (1000 μg/g) (Council, 2006).

Moreover, ICP-MS analysis shows the cleaning process removed most heavy metals: approximately 75-77% of lead (Pb), 90–93% of cadmium (Cd), 60–80% of chromium (Cr), and 50–55% of arsenic (As) (Table S4). Post-cleaning metal levels were significantly reduced (Pb: 1738–1815 μg/kg; Cd: 62-88 μg/kg; Cr: 306-723 μg/kg; As: 1080–1167 μg/kg), all below U.S. EPA hazardous waste limits (5000 μg/kg for Pb, As, Cr; 1000 μg/kg for Cd) ((EPA) 10 NOV 2023) and WHO drinking water guidelines (Environment 21 March 2022). This confirms the cleaned CBs material is safe for membrane fabrication and air filtration without risk of harmful metal

release. Torkashvand et al. (2022) demonstrated the efficacy of recycled CB membranes for water treatment (Torkashvand et al., 2022), and our results confirm this material meets safety standards for commercial use, reduces heavy metal exposure risks, and provides a practical route to repurpose cigarette waste into air purification applications, advancing environmental sustainability goals.

### 3.2. Air filtration performance of PAN/CBs hybrid nanofiber membranes

#### 3.2.1. Effect of PAN-to-CBs weight ratio

*3.2.1.1. Performance comparison.* The filtration performance of electrospun nanofiber membranes is typically evaluated based on two key parameters: particle removal efficiency and airflow resistance (pressure drop). As shown in Fig. 4(a), the membrane fabricated from a total polymer concentration of 8 wt% with a PAN-to-CBs weight ratio of 1:1 was tested for its ability to capture $PM_{1.0}$, $PM_{2.5}$, and $PM_{10}$ at air velocities of 3.3, 5.3, and 7.3 cm/s over time. For particles in the 2.5-10 μm range ($PM_{2.5}$ and $PM_{10}$), the membrane consistently achieved filtration efficiencies exceeding 99.9% across all tested airflow velocities throughout the entire filtration period. In contrast, the efficiency for $PM_{1.0}$ was slightly lower, at 99.45%. Given the excellent performance for larger particulates, it is crucial to explore strategies to further enhance $PM_{1.0}$ capture, particularly by tuning the composition of the membrane.

One promising approach involves systematically varying the weight ratio of PAN to CBs while maintaining a constant total solids concentration (8 wt%). Such an investigation can reveal how compositional changes affect nanofiber morphology, pore architecture, and ultimately, the capture efficiency of ultrafine particles like $PM_{1.0}$. To holistically assess filtration performance, the quality factor was calculated and used as a comparative metric. As illustrated in Fig. 4(b), QF values for $PM_{1.0}$ capture were evaluated across different PAN/CBs weight ratios at the same total concentration (8 wt%) and under varying airflow velocities. Membranes with lower CBs content, specifically PAN/CBs ratios of 1:0 and 1:0.1, exhibited superior QF values (both~0.012 $Pa^{-1}$), primarily due to their significantly lower pressure drops, as confirmed in Fig. 4(c). Fig. 4(c) shows a clear trend: increasing the CBs fraction leads to a progressive

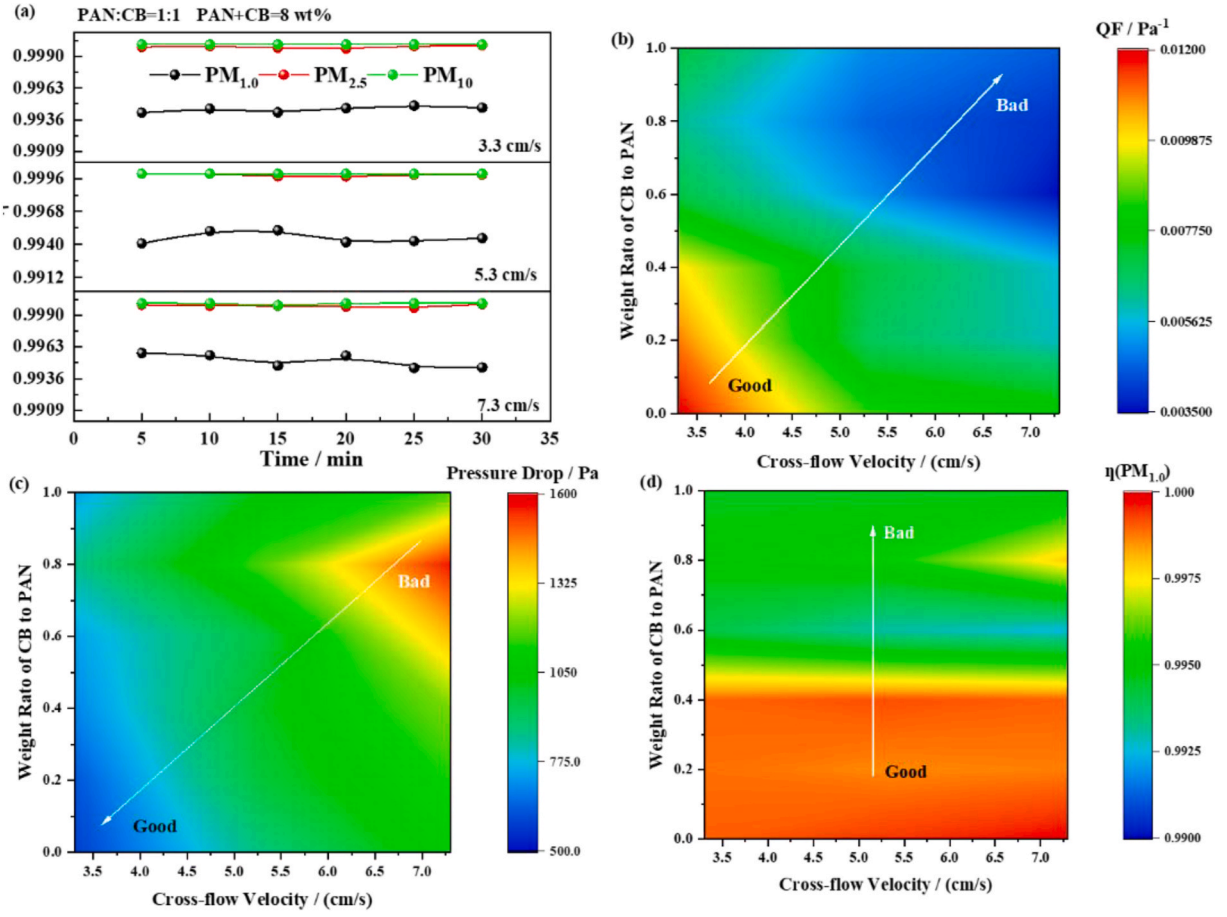
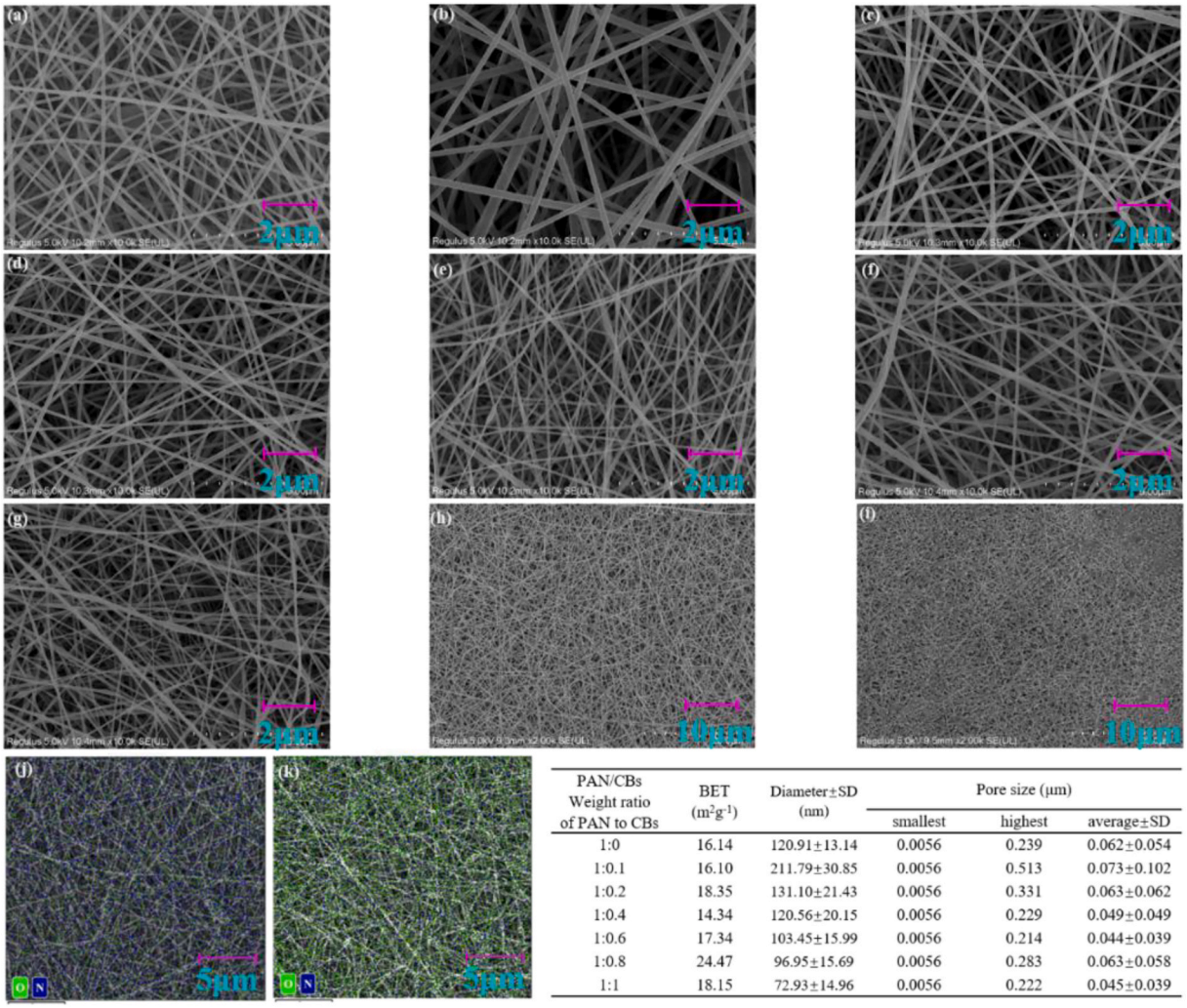



**Fig. 4.** (a) Filtration efficiency of PAN/CBs nanofiber membranes (1:1 wt ratio, 8 wt% total concentration) for $PM_{1.0}$, $PM_{2.5}$, and $PM_{10}$ at airflow velocities of 3.3, 5.3, and 7.3 cm/s, (b) QF for $PM_{1.0}$ capture under the same conditions, (c) Corresponding pressure drop across the membranes at different CBs loadings, and (d) $PM_{1.0}$ filtration efficiency of PAN/CBs membranes (8 wt% total concentration) as a function of PAN-to-CBs weight ratio and airflow velocity.

rise in pressure drop. At an airflow velocity of 3.3 cm/s, the pressure drop increased from 583 Pa (0% CB) to 738 Pa (higher CBs loading); at 5.3 cm/s, it rose from 875 Pa to 1029 Pa; and at 7.3 cm/s, from 1082 Pa to 1162 Pa. This increase is attributed to the formation of agglomerated CBs-derived residues on or within the fiber matrix, which reduces membrane porosity and impedes airflow. Notably, a sharp rise in pressure drop was observed after approximately 30 min of continuous filtration for CBs-containing membranes, likely due to fine particles becoming trapped in the tighter interstitial spaces between fibers. SEM analysis further confirmed that fibers containing CBs exhibit smaller diameters compared to pure PAN fibers, contributing to denser packing and higher flow resistance. Furthermore, as shown in Fig. 4(d), increasing the CBs content also led to a modest but consistent decline in $PM_{1.0}$ filtration efficiency. At airflow velocities of 3.3, 5.3, and 7.3 cm/s, the efficiency decreased from 99.90%, 99.94%, and 99.98% (for 0% CBs) to 99.46%, 99.45%, and 99.46% (at higher CBs loadings), respectively. This reduction is likely caused by morphological disruptions in the nanofibrous network, such as non-uniform fiber distribution, increased surface roughness, or enlarged inter-fiber voids, induced by the incorporation of heterogeneous CBs-derived components. These structural irregularities may compromise the membrane's ability to effectively intercept ultrafine particles via mechanisms like diffusion and interception.

Nevertheless, even at elevated CBs loadings, the filtration efficiency for $PM_{1.0}$ remains above 99.4%, underscoring the robust performance of these hybrid membranes. Critically, this demonstrates that waste cigarette butts can be successfully valorized into functional air filtration materials without significantly compromising performance, offering a sustainable pathway for upcycling a




| PAN/CBs Weight ratio of PAN to CBs | BET ($m^2g^{-1}$) | Diameter±SD (nm) | Pore size (μm) | | |
|---|---|---|---|---|---|
| | | | smallest | highest | average±SD |
| 1:0 | 16.14 | 120.91±13.14 | 0.0056 | 0.239 | 0.062±0.054 |
| 1:0.1 | 16.10 | 211.79±30.85 | 0.0056 | 0.513 | 0.073±0.102 |
| 1:0.2 | 18.35 | 131.10±21.43 | 0.0056 | 0.331 | 0.063±0.062 |
| 1:0.4 | 14.34 | 120.56±20.15 | 0.0056 | 0.229 | 0.049±0.049 |
| 1:0.6 | 17.34 | 103.45±15.99 | 0.0056 | 0.214 | 0.044±0.039 |
| 1:0.8 | 24.47 | 96.95±15.69 | 0.0056 | 0.283 | 0.063±0.058 |
| 1:1 | 18.15 | 72.93±14.96 | 0.0056 | 0.222 | 0.045±0.039 |

**Fig. 5.** SEM images of electrospun PAN/CB nanofiber membranes (8 wt% total concentration) with varying PAN-to-CBs weight ratios: (a) 1:0, (b) 1:0.1, (c) 1:0.2, (d) 1:0.4, (e) 1:0.6, (f) 1:0.8, and (g) 1:1. All images were acquired at a magnification of 10,000×. The corresponding fiber diameter distributions are shown in Fig. S3, whereas the pore-size distributions are presented in Figure S4 (h, i) SEM images (magnification: 2000×) of the membranes after 30 min of filtration testing with incense smoke: (h) PAN/CBs (1:0) and (i) PAN/CBs (1:1). (j, k) EDS elemental mapping of (j) PAN/CBs (1:0) and (k) PAN/CBs (1:1) membranes, highlighting the spatial distribution of nitrogen (from PAN) and oxygen (from cellulose acetate in CBs). Inset (Inverted Table 1): Summary of BET-specific surface area, average fiber diameter, and average pore size for PAN/CBs (8 wt%) membranes as a function of PAN-to-CBs weight ratio.

pervasive environmental pollutant.

*3.2.1.2. Membrane characterization and mechanism analysis.* Following the evaluation of filtration performance in Fig. 4, a comprehensive suite of physicochemical and structural characterizations was carried out to elucidate the underlying mechanisms governing the observed trends in particle capture efficiency, pressure drop, and quality factor. SEM was employed to investigate how the incorporation of CBs-derived components influences the morphology, diameter, and surface architecture of PAN-based nanofibers at a fixed total concentration of 8 wt%, while varying the PAN-to-CBs weight ratio. As shown in Fig. 5(a–g), the electrospun fibers maintain continuous and smooth morphologies across all compositions, from pure PAN (1:0) to a 1:1 PAN/CBs blend, indicating good dispersion of CBs components within the PAN/DMF solution and compatibility during electrospinning. However, a clear reduction in average fiber diameter is observed, decreasing from 120.91 nm (1:0) to 72.93 nm (1:1), as showed in Fig. S3. This thinning effect is consistent with established literature indicating that carbon-based additives incorporation can reduce polymer solution viscosity and enhance charge density on the electrospinning jet, thereby intensifying stretching forces in the electric field and yielding finer fibers (Jo et al., 2024). A similar trend is seen in pore size (Fig. S4), which contracts from 0.62 μm (1:0) to 0.45 μm (1:1), further confirming the densification of the fibrous network with higher CBs content. Post-filtration SEM images, as illustrated in Fig. 5(h–i), reveal that the PAN/CBs (1:1) membrane appears more heavily loaded with captured particles compared to pure PAN (1:0), with visible fiber bending, breakage, and displacement, suggesting greater particle adhesion, as illustrated in Fig. 6. Paradoxically, despite this apparent "dirtier" appearance, its $PM_{1.0}$ filtration efficiency is slightly lower than that of pure PAN. This counterintuitive result underscores a critical principle in filtration science: performance is not solely determined by particle retention but also by airflow dynamics and structural integrity. The denser, thinner-fiber structure of the 1:1 membrane leads to faster clogging and significantly higher resistance to airflow, as corroborated by the elevated pressure drop in Fig. 4(c), making it less energy-efficient despite its high initial capture capacity (Zhang et al., 2019). EDS mapping illustrated in Fig. 5(j–k) provides direct evidence of molecular-level integration between PAN and CBs components. Nitrogen (N), characteristic of PAN's nitrile groups, and oxygen (O), primarily from cellulose acetate in CBs, are co-localized across individual fibers. With increasing CBs content (from 0% to 50%), N intensity decreases while O increases, confirming uniform blending rather than phase separation (Lakshmanan et al., 2021).

Notably, under identical electrospinning conditions, the thickness of all membranes ranges from 48 μm (pure PAN (1:0)) to 45 μm (PAN/CBs (1:1)), with the exception of the 1:0.8 sample (32 μm), as shown in Fig. 7(a). At this intermediate loading, the addition of cellulose acetate from CBs may transiently reduce solution viscosity or alter jet stretching dynamics, yielding finer fibers and/or lower deposition rates. However, this trend was reproducible across ≥5 independent electrospinning batches (Table S5). And, even with substantial incorporation of hydrophilic cellulose acetate, the membranes retain strong hydrophobicity, as evidenced by water contact

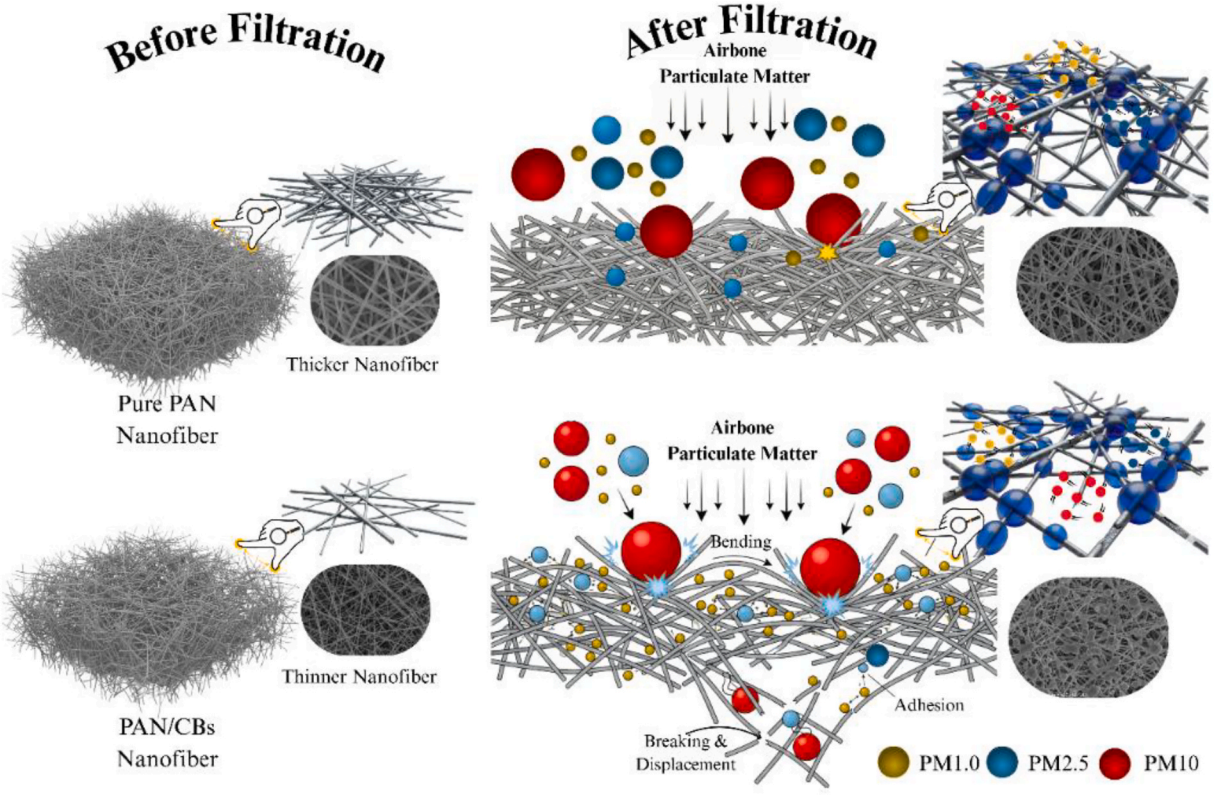



**Fig. 6.** Schematic diagram of the possible mechanism to illustrate the difference of PAN and PAN/CBs membranes in the particle removal efficiency.

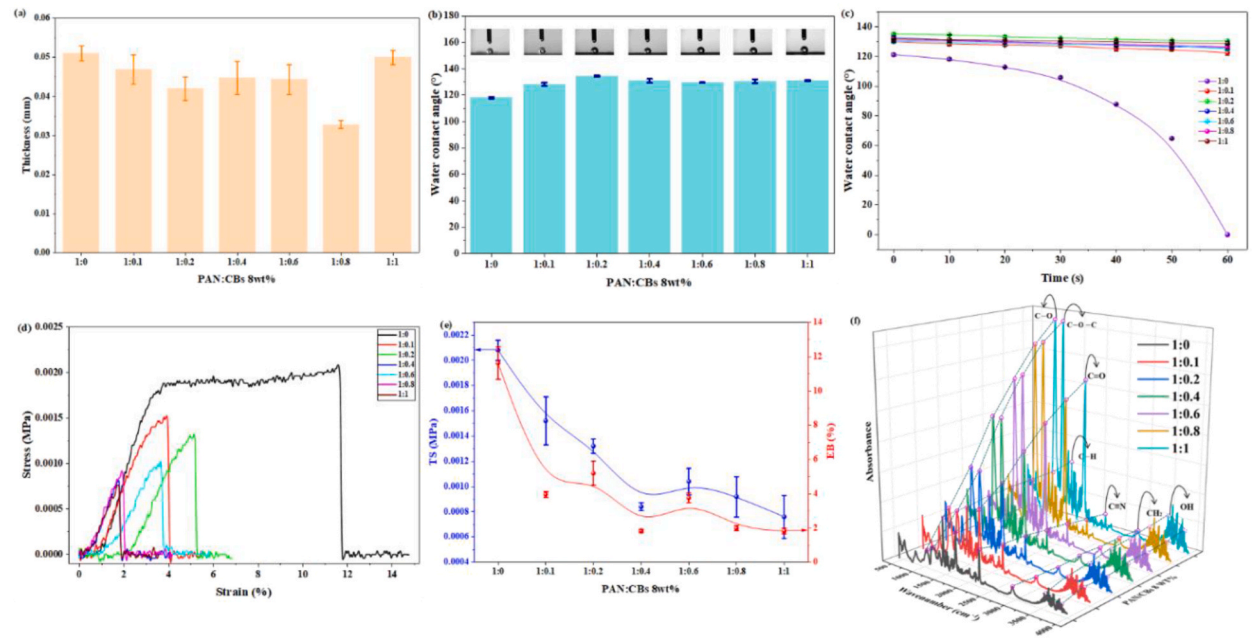



**Fig. 7.** (a) Membrane thickness, (b) water contact angle, (c) water contact angle over 60s, (d) stress-strain curves, (e) tensile stress (TS) and elongation at break (EB), and (f) FT-IR spectra of PAN/CBs 8 wt% total concentration nanofiber membranes under different weight ratio of PAN to CBs.

angles consistently above 120° (Fig. 7(b)). In contrast, pure PAN unexpectedly transitions from hydrophobic (~118°) to hydrophilic (<90°) within 60 s (Fig. 7(c)), likely due to surface hydrolysis of nitrile groups into carboxylic acids during storage or processing. The sustained hydrophobicity in PAN/CBs composites suggests that PAN remains dominant at the fiber surface, effectively shielding the internal hydrophilic CA domains, a feature highly beneficial for air filtration, as it prevents moisture-induced pore clogging under humid conditions. This structural and surface behavior directly influences specific surface area, as quantified by BET analysis (summarized in the inset table of Fig. 5). The addition of CBs increases surface area from 16.14 $m^2/g$ (1:0) to a peak of 24.47 $m^2/g$ at the 1:0.8 ratio, despite the latter membrane being physically thinner, highlighting how nanoscale roughness and reduced fiber diameter can compensate for reduced thickness. Overall, surface area inversely correlates with fiber diameter: finer fibers yield higher surface-to-mass ratios, enhancing potential interaction sites for particle capture. However, mechanical robustness presents a trade-off. Tensile strength declines markedly with CBs loading, from 0.0021 MPa (1:0) to 0.0008 MPa (1:1), due to poor interfacial adhesion, filler agglomeration, and disruption of the continuous PAN network (Fig. 7(d–e)). While low CBs content offers marginal reinforcement, higher loadings induce brittleness, emphasizing the need for compositional optimization or hybrid strategies, such as increased the spinning time during electrospinning or blending with some other materials, to balance filtration performance with mechanical durability (Zhou et al., 2025). Finally, FT-IR spectroscopy illustrated in Fig. 7(f) confirms the chemical coexistence of both components without degradation. The signature nitrile (C ≡ N) stretch of PAN at ~2242 $cm^{-1}$ and $CH_2$ bending at ~2940 $cm^{-1}$ remain present across all samples, though their intensity diminishes with increasing CBs content (Zhou et al., 2025). Concurrently, new peaks emerge, most notably the ester carbonyl (C═O) stretch at ~1748 $cm^{-1}$, along with C–O–C (~1235 $cm^{-1}$), C–O (~1046 $cm^{-1}$), and CH bending (~1374 $cm^{-1}$), all characteristic of cellulose acetate (Kakoria and Sinha-Ray, 2022). The broad band between 3502 and 3908 $cm^{-1}$ corresponds to adsorbed moisture. Crucially, peak positions remain unchanged regardless of composition, indicating no chemical reaction between PAN and CBs during electrospinning. Thus, CBs function not as inert fillers but as functional additives that introduce polar ester groups into the matrix, potentially enhancing electrostatic or dipole-based particle interactions, while preserving the structural backbone of PAN.

Collectively, these characterizations reveal that while CBs incorporation refines fiber morphology, increases surface area, and imparts functional chemical groups, it also introduces trade-offs in pressure drop, mechanical strength, and ultrafine particle capture efficiency. Nevertheless, the PAN/CBs hybrid membranes, particularly at moderate CBs loadings, demonstrate a compelling balance of high filtration performance (>99.4% for $PM_{1.0}$), hydrophobic stability, and waste valorization potential, positioning them as promising candidates for sustainable air filtration technologies.

### *3.2.2. Effect of post-heat treatment*

After investigating the effect of varying the PAN to CBs weight ratio at a fixed 8 wt% total polymer concentration on air filtration performance, the results obtained particularly for $PM_{1.0}$ capture however did not yet meet the desired efficiency targets, despite promising performance for $PM_{2.5}$ and $PM_{10}$. Among the formulations tested, the PAN/CBs 8 wt% membrane with a 1:1 wt ratio exhibited a favorable balance of fine fiber morphology, reduced pore size, and relatively high surface area, making it the most suitable

candidate for further optimization. Consequently, nanofiber membrane was selected to study the impact of post-electrospinning heat treatment at 100, 120, and 140°C on both air filtration performance and membrane morphology, with the aim of enhancing structural integrity and ultrafine particle capture efficiency through controlled thermal stabilization. As illustrated in Fig. 8(a), the filtration performance of the PAN/CBs 1:1 8 wt% at different heat-treatment temperature at air-flow velocity of 5.3 cm/s, over-time to capture $PM_{1.0}$, $PM_{2.5}$, and $PM_{10}$. Through heat-treatment, it can be seen an improved on the particle removal efficiency especially for $PM_{1.0}$ capture from 99.78% at 100°C to 99.88% at 140°C. These promising results however accomplished with a higher airflow resistance (pressure drop in Fig. 8(b)) of all air-flow velocity from 738 Pa to 1002 Pa at 3.3 $cms^{-1}$, 1029 Pa to 1550 Pa at 5.3 $cms^{-1}$, and 1162 Pa to 1995 Pa at 7.3 $cms^{-1}$. Thus, resulting in a lower quality factor (QF) in Fig. 8(c), which balances efficiency against pressure drop.

Morphological analysis of the SEM images in Fig. 9(a–c) reveals that increasing the heat-pressing temperature from ambient (0°C, reference sample shown in Fig. 5(g)) to 140°C progressively induces fiber expansion and coalescence. Consequently, the average fiber diameter increases from 72.93 nm (0°C) to 81.08 nm (100°C), 90.42 nm (120°C), and 90.82 nm (140°C), as illustrated in Fig. S5. Concurrently, the average pore size (Fig. S6) decreases from 0.045 μm (0°C) to 0.038 μm (100°C), 0.034 μm (120°C), and slightly rebounds to 0.037 μm (140°C). This reduction in pore size may enhance particle interception; however, the thicker fibers and denser fibrous network significantly compromise air permeability (Zhang et al., 2020). As evidenced by the post-filtration surface morphology in Fig. 9(d), this structural densification leads to a marked increase in pressure drop without commensurate gains in filtration efficiency, particularly for fine particles such as $PM_{1.0}$. Moreover, the membrane thickness (illustrated in Fig. 9(g)) become flattered after heat pressing treatment from 0.05 mm at 0°C to approximately 0.025 mm across all temperature of 100-140°C, which can be ascribe to an improve in the nanofibers' adhesion and tighten their inter-junction bonding to cause physical fusion among fibers with impeding fiber slippage while restricting their mobility. Collectively, these findings indicate that, under the current processing conditions, heat pressing alone is insufficient to simultaneously achieve high filtration efficiency and low airflow resistance. This underscores the need for complementary strategies, such as optimizing polymer solution concentration, to achieve high-performance, low-resistance air filtration membranes from recycled cigarette butts.

Additionally, EDS elemental mapping of the nanofiber membrane after heat treatment at 120°C and exposure to incense smoke (Fig. 9(e–f)) confirms the capture of a diverse array of combustion byproducts. Detected elements include nitrogen (N), oxygen (O), silicon (Si), sulfur (S), chlorine (Cl), and potassium (K). Notably, potassium serves as a well-established tracer for incense smoke, originating from potassium-based additives such as potassium nitrate used as oxidizers or binders. Sulfur and chlorine likely derive

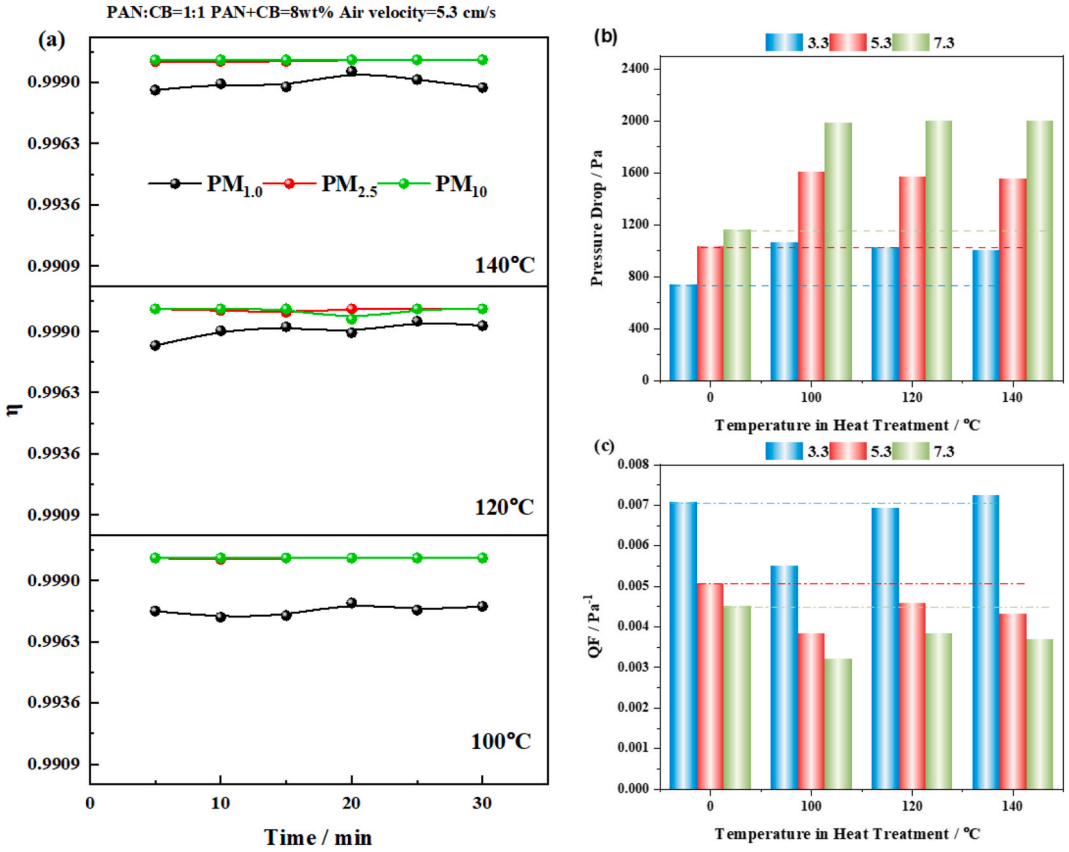



**Fig. 8.** (a) Particle removal efficiency of $PM_{1.0}$, $PM_{2.5}$, and $PM_{10}$ at 5.3 cm/s air-flow velocity, (b) pressure drop, and (c) quality factor under different post-heat treatment temperature of PAN/CBs 1:1 8 wt% total concentration.

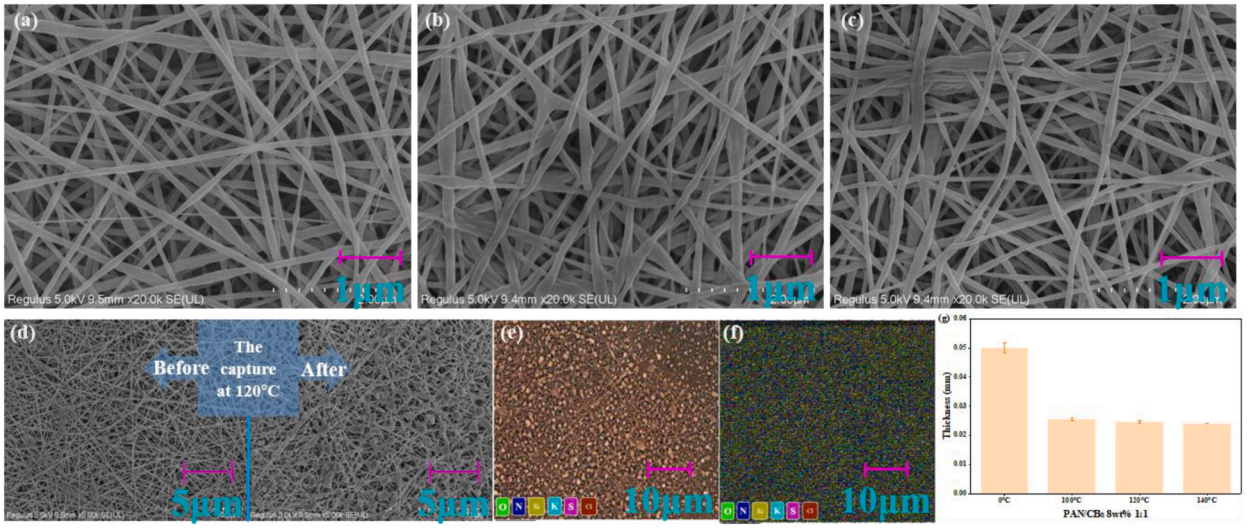



**Fig. 9.** SEM images of PAN/CBs 1:1 8 wt% nanofiber under different post-heat treatment of (a) 100°C, (b) 120°C, (c) 140°C. The corresponding fiber diameter distributions are shown in Fig. S5, whereas the pore-size distributions are presented in Fig. S6. (d) SEM images of PAN/CBs 1:1 8 wt% nanofiber at 120°C before and after the capture of incense smoke for 30 min, (e) original EDS mapping image, (f) component revealed attract to the fiber surface EDS mapping image of PAN/CBs 1:1 8 wt% at 120°C after filtering incense smoke, and (g) the membrane thickness of PAN/CBs 1:1 8 wt% under heat pressing treatment.

from dyes, fragrances, or impurities that release sulfurous or chlorinated compounds upon combustion. Silicon is probably associated with inorganic fillers (e.g., talc or silica) commonly incorporated into incense sticks (Mannix et al., 1996). Meanwhile, N and O signals arise both from the membrane matrix (comprising PAN and cellulose acetate from cigarette butts) and from oxygen- and nitrogen-containing organic pollutants (e.g., VOCs and aromatic compounds) present in the smoke. The detection of this broad spectrum of elements on the membrane surface demonstrates its capability not only to capture particulate matter but also to adsorb complex chemical constituents generated during real-world incense burning, highlighting its promising potential for practical indoor air purification applications.

### *3.2.3. Effects of total weight ratio of PAN to CBs*

*3.2.3.1. Performance comparison.* Following the initial investigation into the effect of varying the PAN-to-CBs weight ratio at a fixed total polymer concentration of 8 wt%, it was observed that while the particle removal efficiency for larger particulates ($PM_{2.5}$ and $PM_{10}$) remained excellent, $PM_{1.0}$ capture by PAN/CBs (1:1) membranes fell slightly short of the performance achieved by pure PAN, despite still exceeding 99.4%. To further understand how both total polymer concentration and CBs incorporation jointly influence membrane microstructure and filtration behavior, an expanded series of formulations was prepared and evaluated: PAN/CBs (1:0 and 1:1) at total concentrations of 8, 10, and 14 wt%. This systematic approach allows for a direct assessment of how increasing polymer content, either with or without CBs, affects fiber morphology, packing density, and the critical trade-off between particle capture efficiency and airflow resistance.

Fig. 10(a) presents the $PM_{1.0}$ filtration efficiencies of pure PAN (1:0) and PAN/CBs (1:1) nanofiber membranes across the three polymer concentrations and at airflow velocities of 3.3, 5.3, and 7.3 cm/s. At 8 wt%, pure PAN consistently achieves >99.9% efficiency, whereas PAN/CBs (1:1) shows a modest reduction (99.45-99.6%), likely due to structural heterogeneities introduced by the recycled CBs material. However, when the total concentration is increased to 10 wt%, the PAN/CBs (1:1) membrane not only matches but slightly exceeds pure PAN, reaching 99.97-99.99% efficiency, even at the highest airflow velocity. This indicates that a higher polymer concentration promotes the formation of a denser, more cohesive fibrous network that effectively compensates for the disruptive effects of CBs incorporation. At 14 wt%, PAN/CBs (1:1) maintains robust performance (≥99.83%), while pure PAN exhibits a slight decline at 7.3 cm/s (99.62%), possibly due to excessive thickness or reduced porosity that leads to non-uniform airflow and particle bypass. This behavior may be attributed to the residual static charges retained from the electrospinning process, especially when the CBs is present. During electrospinning, the high voltage and rapid solvent drying lock electric charges into the solidified fibers, and due to CBs fibers are mainly composed of cellulose acetate, a polar material that naturally holds onto static charges well, they help stabilize these electric fields within the composite nanofibers during fiber formation. These electrostatic forces attract and capture fine particles through direct electrical attraction, at higher airflow speed (Kim et al., 2024). This helps compensate for the shorter contact time between PM particles and the filter when air moves faster. Similar mechanisms were also found in the scientific literature (Jiang et al., 2024; Scaffaro et al., 2026). Collectively, these results demonstrate that CBs incorporation becomes advantageous at higher polymer concentrations, with the 14 wt% PAN/CBs (1:1) membrane delivering near-complete $PM_{1.0}$ removal across all conditions, highlighting its potential as a high-performance, sustainable air filter in which waste-derived CBs enhance rather than

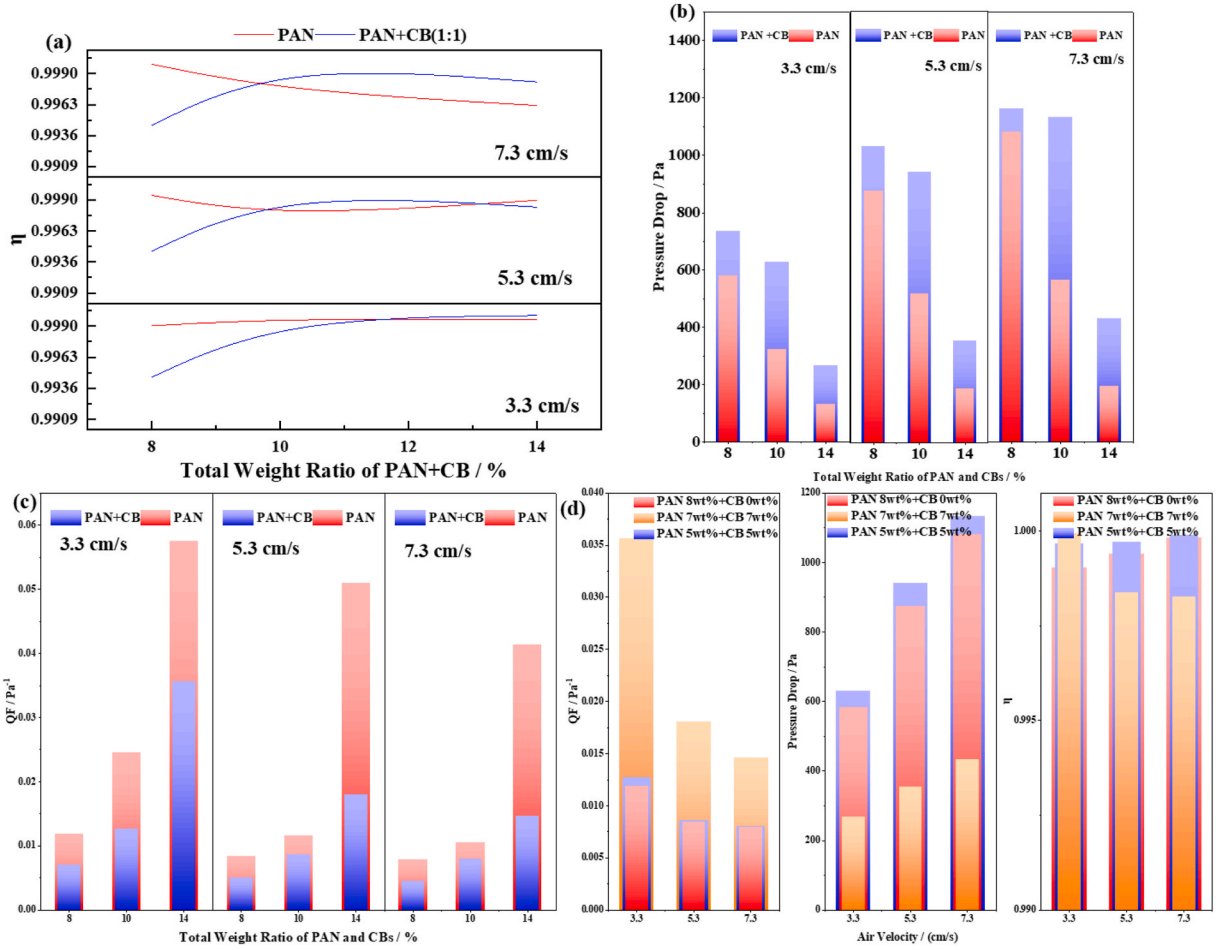



**Fig. 10.** (a) Particle removal efficiency, (b) pressure drop, and (c) quality factor (QF) for $PM_{1.0}$ capture by electrospun pure PAN (1:0) and PAN/CBs (1:1) nanofiber membranes at total polymer concentrations of 8, 10, and 14 wt%, evaluated under airflow velocities of 3.3, 5.3, and 7.3 cm/s. (d) Comparison of air filtration performance between PAN/CBs (1:1) membranes at 10 wt% and 14 wt% total polymer concentration and pure PAN (1:0) at 8 wt%.

compromise functionality when properly formulated.

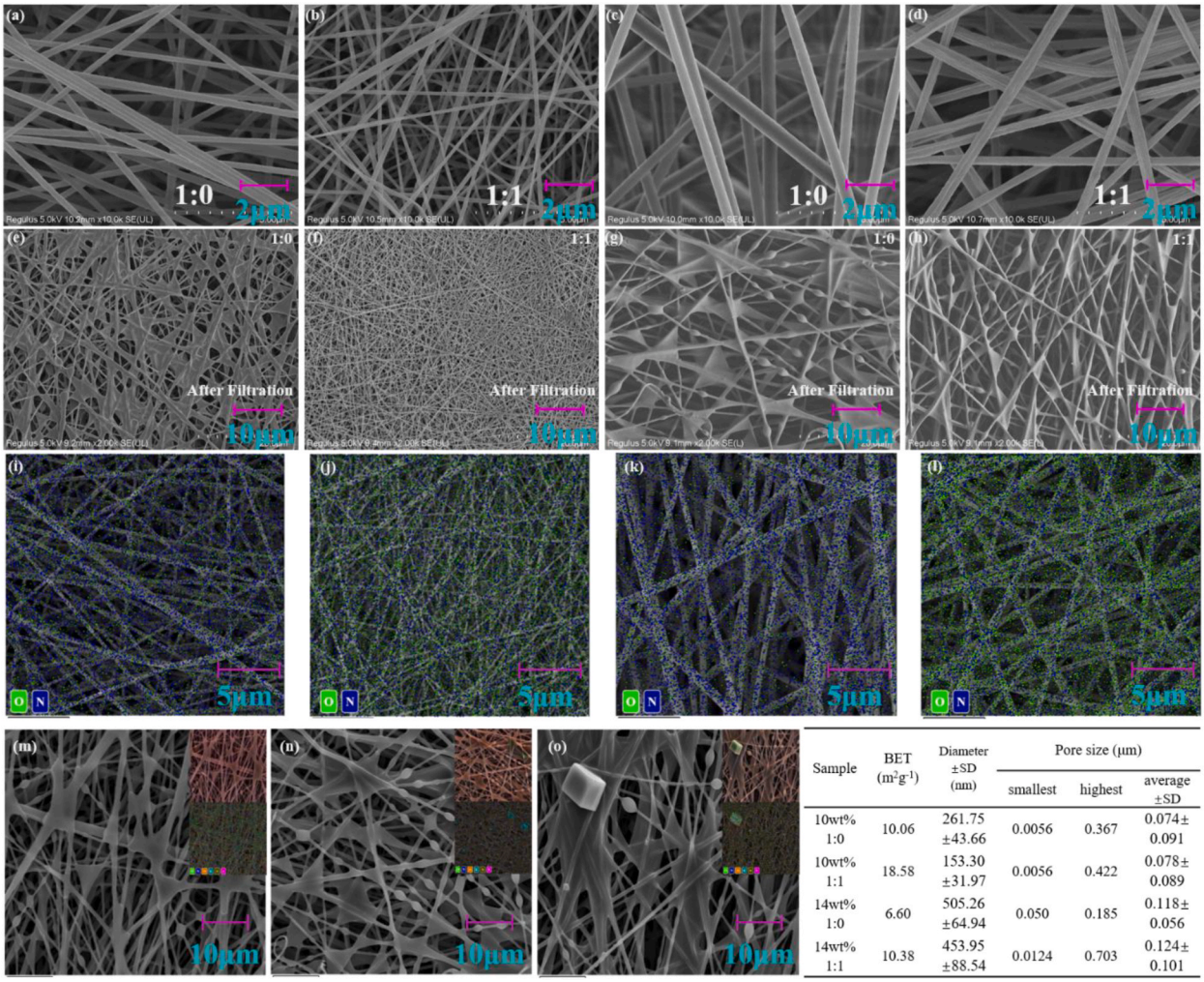
Pressure drop measurements are shown in Fig. 10(b). Remarkably, despite the substantial improvement in filtration efficiency, increasing the total polymer concentration from 8 to 14 wt% leads to a significant reduction in pressure drop for both membrane types across all airflow velocities. For pure PAN, the pressure drop at 7.3 cm/s decreases dramatically from 1082 Pa (8 wt%) to 566 Pa (10 wt %) and further to just 197 Pa (14 wt%). Similarly, for PAN/CB (1:1), it drops from 1162 Pa (8 wt%) to 1133 Pa (10 wt%) and then to 434 Pa (14 wt%). This counterintuitive trend suggests that higher polymer concentrations yield more open, uniform, or better-organized fibrous architectures during electrospinning, enhancing permeability even as more material is deposited. Nevertheless, PAN/CBs (1:1) membranes consistently exhibit higher pressure drops than their pure PAN counterparts at equivalent concentrations, confirming that CBs incorporation increases structural density or reduces effective porosity. Importantly, the 14 wt% PAN/CBs membrane still achieves a relatively low pressure drop (269 Pa at 3.3 cm/s; 434 Pa at 7.3 cm/s) while maintaining >99.8% $PM_{1.0}$ efficiency, demonstrating an excellent balance between high capture performance and pressure drop.

The quality factor (QF), which integrates efficiency and pressure drop into a single metric of overall filtration performance, is summarized in Fig. 10(c). In all cases, QF improves markedly with increasing polymer concentration. For pure PAN at 3.3 cm/s, QF rises from 0.0119 $Pa^{-1}$ (8 wt%) to 0.0246 $Pa^{-1}$ (10 wt%) and sharply to 0.0575 $Pa^{-1}$ (14 wt%). PAN/CB (1:1) shows a parallel trend, increasing from 0.0071 $Pa^{-1}$ (8 wt%) to 0.0127 $Pa^{-1}$ (10 wt%) and 0.0356 $Pa^{-1}$ (14 wt%). This enhancement persists even at 7.3 cm/s, where the 14 wt% PAN and PAN/CBs membranes achieve QFs of 0.0413 $Pa^{-1}$ and 0.0147 $Pa^{-1}$, respectively, both substantially higher than their 8 wt% counterparts. Although PAN/CBs nanofiber membranes consistently exhibit lower QF values than pure PAN at matched concentrations (due to higher pressure drops), their performance improves significantly with increased polymer loading. Notably, the 14 wt% PAN/CBs nanofiber membrane attains a respectable QF of 0.0356 $Pa^{-1}$ at 3.3 cm/s, confirming that recycled CBs can be integrated into high-efficiency filters without severely compromising the efficiency-resistance balance.

Finally, Fig. 10(d) directly compares the filtration performance of PAN/CBs (1:1) nanofiber membranes at 10 wt% and 14 wt% with

that of pure PAN at 8 wt%. The results reveal a key insight: by fixing the PAN:CBs ratio at 1:1 and simply increasing the total polymer concentration, the composite membrane can match or even surpass the performance of a pure PAN nanofiber membrane containing twice as much PAN. This is particularly significant given that PAN is a high-performance but costly polymer, whereas CBs are abundant, low-cost waste materials. While CBs alone lack sufficient filtration capability, blending them equally with PAN and processing at higher concentrations (10-14 wt%) yields membranes with fine, densely packed fibers and optimized pore structures that efficiently capture $PM_{1.0}$ while maintaining acceptable airflow. Thus, replacing half of the PAN with valorized cigarette butt waste does not degrade (indeed, under optimal conditions, it can enhance) overall filtration performance, offering a compelling strategy for sustainable, cost-effective air purification.

*3.2.3.2. Membrane characterization and mechanism analysis.* Following the observation of enhanced air filtration performance for PAN/CBs nanofiber membranes at higher polymer concentrations (Fig. 10), a comprehensive set of morphological, surface, and chemical characterizations was carried out to elucidate the underlying mechanisms. SEM images in Fig. 11(a–d) reveal that increasing the total polymer concentration from 8 wt% to 14 wt% leads to a pronounced thickening of fibers: the average diameter of pure PAN (1:0) rises from 121 nm (8 wt%, Fig. 5(a)) to 261.75 nm (10 wt%, Fig. 11(a)) and further to 505.26 nm (14 wt%, Fig. 11(c)), while PAN/CB (1:1) fibers increase from 73 nm (8 wt%, Fig. 5(g)) to 153.30 nm (10 wt%, Fig. 11(b)) and 453.95 nm (14 wt%, Fig. 11(d)),




| Sample | BET ($m^2g^{-1}$) | Diameter ±SD (nm) | Pore size (μm) | | |
|---|---|---|---|---|---|
| | | | smallest | highest | average ±SD |
| 10wt% 1:0 | 10.06 | 261.75 ±43.66 | 0.0056 | 0.367 | 0.074± 0.091 |
| 10wt% 1:1 | 18.58 | 153.30 ±31.97 | 0.0056 | 0.422 | 0.078± 0.089 |
| 14wt% 1:0 | 6.60 | 505.26 ±64.94 | 0.050 | 0.185 | 0.118± 0.056 |
| 14wt% 1:1 | 10.38 | 453.95 ±88.54 | 0.0124 | 0.703 | 0.124± 0.101 |

**Fig. 11.** SEM images of PAN/CBs nanofiber membranes at total polymer concentrations of 10 wt% and 14 wt% with PAN:CBs weight ratios of (a) 1:0, (b) 1:1, (c) 1:0, and (d) 1:1, respectively. All images are acquired at a magnification of 10,000×. The corresponding fiber diameter distributions are shown in Fig. S7, whereas the pore-size distributions are presented in Figure S8 (e-h) SEM images of the same membranes after 30 min of incense smoke filtration, imaged at 2000× magnification. (i-l) EDS elemental mapping of the as-spun membranes corresponding to (i) 10 wt% (1:0), (j) 10 wt% (1:1), (k) 14 wt% (1:0), and (l) 14 wt% (1:1). (m–o) SEM-EDS mapping of a pure PAN (1:0) membrane at 14 wt% after filtrations under airflow velocities of (m) 3.3 cm/s, (n) 5.3 cm/s, and (o) 7.3 cm/s, showing particulate deposition patterns. Inverted Table 2. BET-specific surface area, along with measured average fiber diameter and pore size, for PAN (1:0) and PAN/CBs (1:1) nanofiber membranes at 10 wt% and 14 wt% total polymer concentrations.

whereas their fiber diameter distributions are shown in Fig. S7. Concurrently, pore sizes expand from 0.062 μm (pure PAN) and 0.045 μm (PAN/CBs) at 8 wt% (Fig. 5(a–g)) to 0.118 μm and 0.124 μm at 14 wt% (Fig. 11(c–d)), whereas their pore-size distributions are shown in Fig. S8, and membrane thickness increases, with pure PAN growing from 51 μm (Fig. 7(a)) to 69 μm (Fig. 12(a)). Although larger pores might intuitively compromise filtration, the combined effect of greater thickness and denser fiber packing enables depth filtration, allowing particles to be captured throughout the membrane bulk rather than solely on the surface, thereby improving $PM_{1.0}$ capture efficiency without a proportional rise in pressure drop. Notably, even though PAN/CBs membranes are thinner than their pure PAN counterparts at equivalent concentrations, they maintain high filtration efficacy, especially at 14 wt%, owing to their relatively finer fibers that form a tighter, more tortuous network capable of compensating for reduced thickness. Post-filtration SEM images (Fig. 11(e–h)) confirm that both membrane types retain structural integrity after exposure to incense smoke, with PAN/CBs membranes exhibiting denser particle deposition, a feature consistent with earlier reports that CBs incorporation promotes a more compact fibrous architecture. EDS mapping of membranes at 10 wt% and 14 wt% (Fig. 11(i–l)) further supports this: nitrogen signals (from PAN's nitrile groups) dominate in pure PAN, while oxygen content (originating from cellulose acetate in CBs) is significantly elevated in PAN/CBs composites. Importantly, this elemental trend persists even at higher polymer loadings, indicating uniform dispersion and chemical stability of CBs within the fibers. Moreover, SEM-EDS analysis of pure PAN (14 wt%) after filtration under varying airflow velocities (Fig. 11(m–o)) demonstrates a clear dependence of particle loading on flow rate: minimal deposition occurs at 3.3 cm/s, moderate accumulation at 5.3 cm/s, and heavy particle adhesion at 7.3 cm/s. This behavior arises because faster airflow increases the number of particles, fiber collisions per second, enhancing capture probability; however, excessive accumulation may accelerate pore clogging, leading to higher pressure drop and potentially shortened filter lifespan (Cui et al., 2021). The absence of significant particle rebound suggests that inertial impaction and interception remain highly effective across all tested velocities.

Complementing these morphological insights, BET surface area measurements (Inverted Table 2 in Fig. 11) show that PAN/CBs (1:1) membranes possess higher specific surface areas, 18.58 $m^2/g$ at 10 wt% and 10.38 $m^2/g$ at 14 wt%, compared to pure PAN (10.06 $m^2/g$ and 6.60 $m^2/g$, respectively). This indicates that despite fiber coarsening with increasing concentration, which typically reduces surface area, the inclusion of CBs introduces micro-roughness or internal porosity that partially offsets this loss. Nevertheless, the overall decline in surface area with higher polymer content reflects the dominant influence of fiber thickening. Surface wettability, assessed via water contact angle (WCA, Fig. 12(b)), reveals strong hydrophobicity across all formulations, with values ranging from 123° to 130°. Droplets remain stable over 60 s (Fig. 12(c)), confirming robust water repellency and high chain crystallinity that limits moisture penetration. This stems from PAN's preferential migration to the fiber surface during electrospinning, effectively encapsulating the hydrophilic hydroxyl and ester groups of cellulose acetate within the fiber core. A slight WCA reduction for PAN/CBs (123.2°) versus pure PAN (130.3°) at 14 wt% may reflect minor surface exposure of CBs components at very high concentrations (Ding et al., 2006), yet the membrane remains distinctly hydrophobic. Additionally, the subtle increase in WCA with rising polymer concentration suggests that thicker, smoother fibers further enhance surface non-wettability.

Mechanical testing (Fig. 12(d and e)) uncovers a nuanced trade-off between strength and flexibility. While pure PAN exhibits superior ductility, elongating by 40.86% before failure compared to only 14.26% for PAN/CB, indicating greater flexibility and energy absorption capacity, an unexpected reversal occurs at 14 wt%: PAN/CB (1:1) achieves a tensile strength of 0.0036 MPa, over three

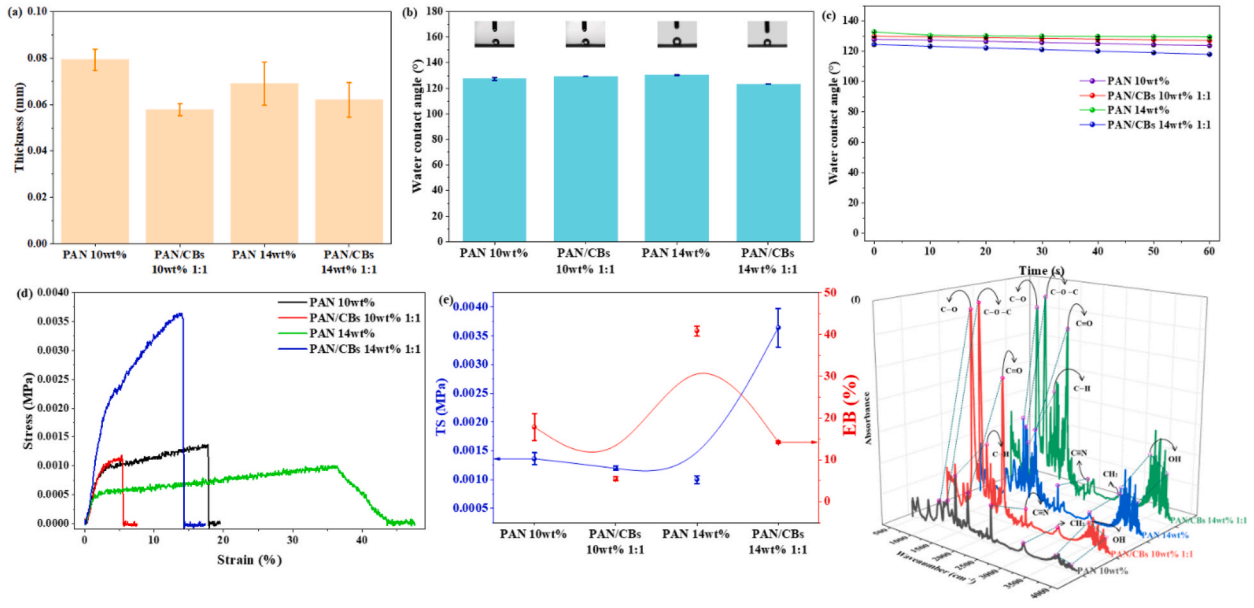



**Fig. 12.** (a) Membrane thickness, (b) water contact angle, (c) water contact angle over 60s, (d) stress-strain curves, (e) tensile stress (TS) and elongation at break (EB), and (f) FT-IR spectra of PAN/CBs 10 wt% and 14 wt% total concentration nanofiber membranes under different weight ratio of PAN to CBs of (1:0 and 1:1).

times higher than pure PAN (0.001 MPa). This implies that at high polymer concentrations, CBs may facilitate the formation of a more cohesive or densely packed fibrous network that better resists fracture under load (Han et al., 2007; Jo et al., 2024). Thus, although PAN/CBs membranes are stiffer and less stretchable, their enhanced strength at elevated concentrations renders them mechanically suitable for practical air filtration applications where handling durability matters.

Finally, FT-IR analysis (Fig. 12(f)) confirms the physical nature of the PAN/CB blend. Characteristic peaks of PAN, including C–H stretching at ~2939 $cm^{-1}$ and C ≡ N stretching at ~2242 $cm^{-1}$, are present in all samples, while PAN/CBs membranes additionally display peaks from cellulose acetate at ~1748 $cm^{-1}$ (C═O), ~1234 $cm^{-1}$ (C–O–C), ~1046 $cm^{-1}$ (C–O), and ~1374 $cm^{-1}$ (C–H), along with broad O–H bands (3446-3948 $cm^{-1}$) from adsorbed water. Crucially, no peak shifts are observed across 8 wt% (Figs. 7(f)), 10 wt%, and 14 wt% formulations (Fig. 12(f)), demonstrating that PAN and CBs do not undergo chemical reaction during electrospinning but instead form a homogeneous physical mixture. Consequently, tuning the total polymer concentration modulates only the relative abundance of each component, not their intrinsic chemistry, enabling precise optimization of filtration performance while preserving the fundamental material structure.

### 3.2.4. CBs as one sustainable additive to enhance PAN membrane performance in air filtration

*3.2.4.1. Performance comparison.* Following the evaluation of the air filtration performance of pure PAN and PAN/CB (1:1) nanofiber membranes electrospun from 10 wt% and 14 wt% total polymer solutions, a distinct enhancement in filtration efficiency alongside a reduction in pressure drop was observed as the polymer concentration increased. Notably, porosity measurements reveal that the pure PAN nanofiber membranes exhibit a slight decrease in porosity when the polymer concentration increases from 10 wt% (77.60 ± 4.21%) to 14 wt% (74.63 ± 1.25%), as detailed in Table S5. This indicates that the superior filtration performance at 14 wt% is not attributed to increased void space, but rather to a more optimized fibrous network. At the higher concentration, the fibers exhibit greater uniformity and fewer structural defects, forming well-defined air channels that reduce airflow resistance while maintaining robust particle capture capability. This optimal balance between filtration efficiency and air permeability is further corroborated by the maximum QF achieved at 14 wt%.

This observation motivates a systematic investigation into the electrospinnability limits of pure PAN and PAN/CBs blends, with CBs-derived material incorporated as a functional additive. To this end, PAN concentrations are fixed at 8 wt%, 10 wt%, and 14 wt%, and CBs are added to achieve a total polymer concentration of 18 wt%, the highest concentration attainable without needle clogging or disruption of continuous fiber formation under the given processing conditions. This approach enables maximized utilization of recycled CBs while remaining within the electrospinnable window defined by pure PAN. Fig. 13 presents the quality factor results, which reveal a distinct trend in how CBs incorporation influences filtration performance relative to pure PAN membranes. While, their filtration efficiency and pressure drop can be seen in Figs. S9 and S10. Among all samples, the nanofiber membrane fabricated from pure PAN at 14 wt% exhibits the highest overall filtration performance across all tested face velocities (3.3, 5.3, and 7.3 cm/s), with QF values of 0.0575 $Pa^{-1}$, 0.05097 $Pa^{-1}$, and 0.04131 $Pa^{-1}$, respectively. This superior performance is supported by post-filtration SEM-EDS images (Fig. 11(m–o)), which show that most captured particles are retained on the topmost layer of the fibrous network, indicative of efficient surface filtration with minimal penetration into the membrane depth. Importantly, the addition of CBs significantly enhances the performance of membranes formulated with lower PAN content (≤10 wt%). At an air velocity of 3.3 cm/s, the PAN 8 wt% + CBs 8 wt% and PAN 10 wt% + CBs 6 wt% blends achieve QF values of 0.05398 $Pa^{-1}$ and 0.05576 $Pa^{-1}$, respectively, values approaching that of pure PAN 14 wt%. This indicates that CBs can effectively compensate for the lower polymer concentration, enabling near-optimal filtration performance under mild operating conditions (Fig. 13(a)). At medium (5.3 cm/s) and high (7.3 cm/s) air velocities, CBs-containing membranes consistently outperform their pure PAN counterparts at equivalent PAN concentrations (Fig. 13(b–c)).

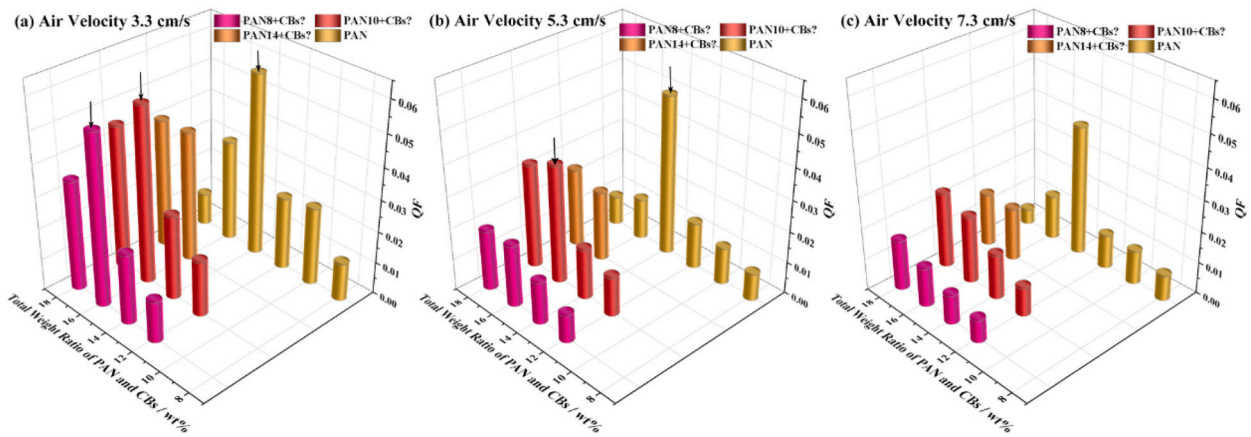
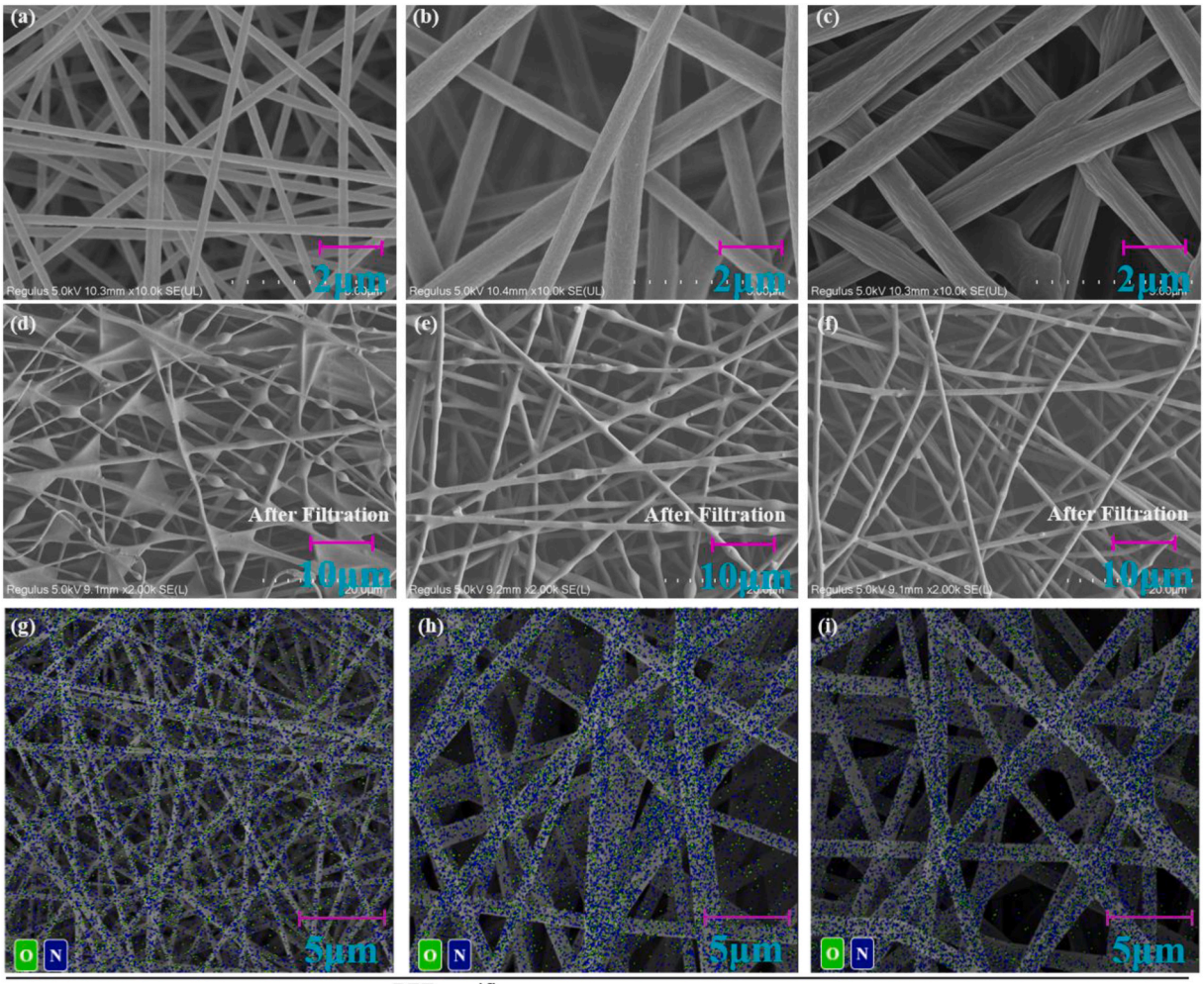



**Fig. 13.** The QF of pure PAN at the concentration from 8 to 18 wt% and fixed PAN with the concentration of 8, 10, and 14 wt% with rising the total solids concentration (PAN + CBs) up to 18 wt% by adding CBs as additives under different air-flow velocity of (a) 3.3 cm/s, (b) 5.3 cm/s, and (c) 7.3 cm/s.

Specifically, PAN 8 wt% + CBs > pure PAN 8 wt%, and PAN 10 wt% + CBs > pure PAN 10 wt%. This enhancement is likely attributed to the finer fiber morphology and modified surface chemistry introduced by CBs, which facilitate more effective particle capture without excessive increase in pressure drop. Nevertheless, none of the CBs-containing membranes surpass the performance of pure PAN 14 wt%. Notably, the PAN 14 wt% + CBs blend exhibits inferior QF compared to pure PAN 14 wt%, suggesting that at high polymer loadings, the presence of CBs disrupts the optimal fiber packing density, uniformity, or inter-fiber architecture responsible for the exceptional performance of the pure PAN system. Therefore, CBs act as a performance-enhancing additive only when the PAN content is moderate or low (≤10 wt%). Under these conditions, blending with waste-derived CBs enables the fabrication of high-efficiency air filters using reduced amounts of virgin PAN, with minimal compromise in filtration performance, particularly under typical indoor air filtration conditions. This strategy offers a sustainable pathway to valorize post-consumer cigarette waste while maintaining competitive membrane functionality.

*3.2.4.2. Membrane characterization and mechanism analysis.* Polymer concentration, solution viscosity, and molecular weight are key

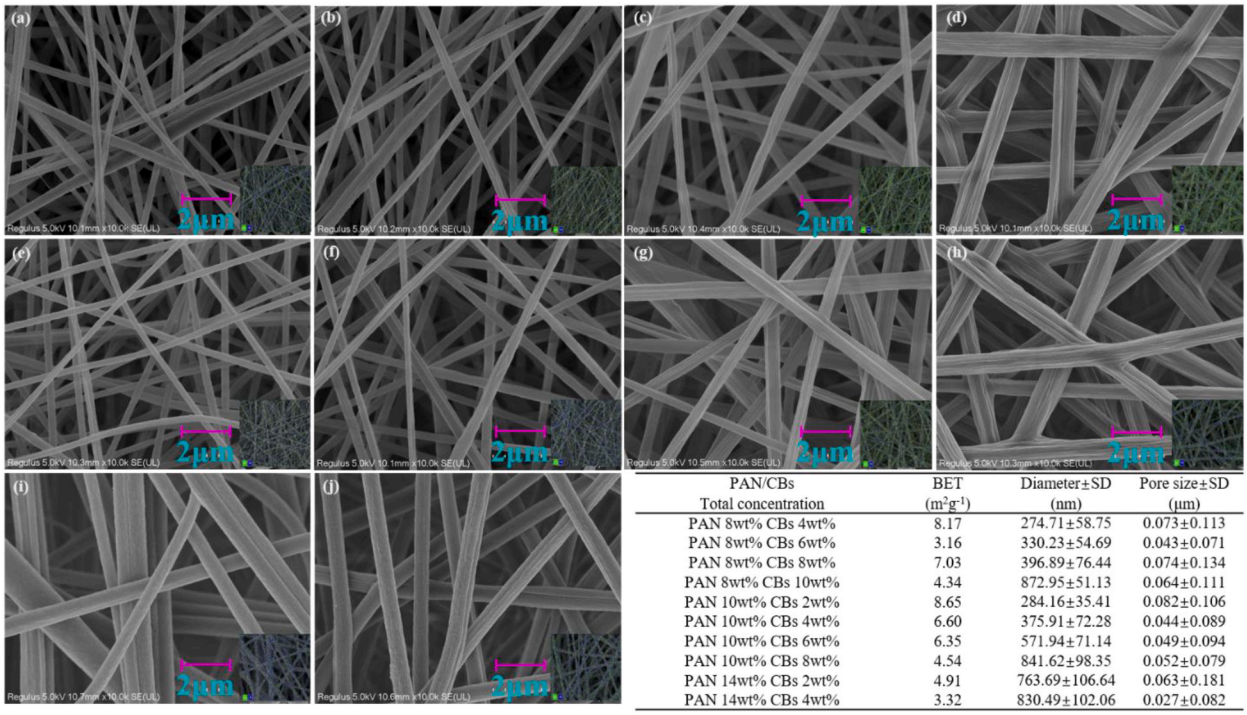



| PAN total concentration | BET specific surface area ($m^2g^{-1}$) | Diameter±SD (nm) | Pore size±SD (μm) |
|---|---|---|---|
| PAN 12wt% | 8.97 | 286.29±28.08 | 0.040±0.079 |
| PAN 16wt% | 5.33 | 908.49±145.21 | 0.203±0.215 |
| PAN 18wt% | 2.88 | 1035.16±126.53 | 0.136±0.11 |

**Fig. 14.** SEM images of PAN nanofiber membranes under different total polymer concentration of (a) 12 wt%, (b) 16 wt%, and (c) 18 wt%. Magnitude is 10000. The corresponding fiber diameter distributions are shown in Fig. S11, whereas the pore-size distributions are presented in Fig. S12. SEM images after the capture of incense smoke for 30 min of PAN nanofiber membranes corresponding to (d) 12 wt%, (e) 16 wt%, and (f) 18 wt%. Magnitude is 2000. EDS mapping analysis of PAN nanofiber membranes corresponding to (g) 12 wt%, (h) 16 wt%, and (i) 18 wt%. Inverted Table 3. Specific Surface Area (BET), and evaluation of the measured average diameter and pore size of pure PAN nanofiber membranes at 12 wt%, 16 wt% and 18 wt% total concentration.

determinants of electrospinnability, as they directly govern the viscoelastic properties of the precursor solution. In general, for a given polymer system, an increase in solution concentration leads to higher viscosity, which typically results in larger fiber diameters and correspondingly enlarged inter-fiber pore sizes. Conversely, lower concentrations yield finer fibers due to reduced chain entanglement and weaker resistance to jet elongation during electrospinning. Fig. 14(a–c) presents SEM images of electrospun polyacrylonitrile (PAN) nanofibers fabricated from solutions with polymer concentrations ranging from 12 wt%, 16 wt%, and 18 wt%, while 8 wt%, 10 wt%, and 14 wt%, can be observed from (Figure 5(a), (11a), and (11c), respectively). All samples exhibit uniform, smooth, and bead-free morphologies, indicating that the selected PAN molecular weight, combined with the applied electrospinning parameters (15 kV voltage, 0.008 mL/h flow rate), is sufficient to stabilize the Taylor cone even under relatively dilute conditions. The average fiber diameter (Fig. S11) increases monotonically with concentration: 120.91 nm (8 wt%), 261.75 nm (10 wt%), 286.29 nm (12 wt%), 505.26 nm (14 wt%), 908.49 nm (16 wt%), and 1035.16 nm (18 wt%), as illustrated in Inverted Table (1-3). This trend is consistent with the expected rise in solution viscosity, which suppresses jet thinning and promotes thicker fiber deposition. Similarly, the average pore size (Fig. S12) expands from 0.062 μm at 8 wt% to 0.136 μm at 18 wt%, reflecting a more open and loosely packed fibrous network at higher concentrations. However, membrane thickness does not follow a linear trend: it increases from 0.0824 mm (12 wt%) to 0.099 mm (16 wt%), but then decreases to 0.0673 mm at 18 wt%. This non-monotonic behavior suggests that while moderate polymer loading enhances fiber deposition and thus membrane thickness, excessively high viscosity at 18 wt% impedes stable jet formation and uniform fiber spreading. The resulting poor spinnability leads to reduced fiber delivery to the collector, ultimately yielding a thinner membrane despite the higher polymer content, a phenomenon also reported in highly viscous systems. Inverted Table 3 quantifies the correlation between fiber diameter, pore size, and concentration for 12, 16, and 18 wt% PAN membranes. The progressive increase in fiber diameter contributes to reduced airflow resistance, which can enhance particle capture efficiency by promoting inertial impaction and interception mechanisms. This observation aligns with prior studies demonstrating that moderately thick fibers improve filtration performance by balancing permeability and capture capacity (Muadtrap et al., 2025; Zhang et al., 2023b).

Beyond the 18 wt% threshold, further increasing PAN concentration to 20 wt% (Fig. S13a) exceeds the critical concentration for stable electrospinning under the current conditions. At this point, rapid solvent evaporation at the needle tip causes premature polymer solidification, leading to nozzle clogging and discontinuous fiber formation, consistent with established failure modes in over-concentrated systems (Haider et al., 2018). Conversely, concentrations below 8 wt% (Fig. S13b) fail to achieve sufficient chain entanglement, resulting in electrospraying rather than fiber formation, as droplets dominate the ejected jet due to insufficient viscoelastic recovery (Shenoy et al., 2005).




| PAN/CBs Total concentration | BET ($m^2g^{-1}$) | Diameter±SD (nm) | Pore size±SD (μm) |
|---|---|---|---|
| PAN 8wt% CBs 4wt% | 8.17 | 274.71±58.75 | 0.073±0.113 |
| PAN 8wt% CBs 6wt% | 3.16 | 330.23±54.69 | 0.043±0.071 |
| PAN 8wt% CBs 8wt% | 7.03 | 396.89±76.44 | 0.074±0.134 |
| PAN 8wt% CBs 10wt% | 4.34 | 872.95±51.13 | 0.064±0.111 |
| PAN 10wt% CBs 2wt% | 8.65 | 284.16±35.41 | 0.082±0.106 |
| PAN 10wt% CBs 4wt% | 6.60 | 375.91±72.28 | 0.044±0.089 |
| PAN 10wt% CBs 6wt% | 6.35 | 571.94±71.14 | 0.049±0.094 |
| PAN 10wt% CBs 8wt% | 4.54 | 841.62±98.35 | 0.052±0.079 |
| PAN 14wt% CBs 2wt% | 4.91 | 763.69±106.64 | 0.063±0.181 |
| PAN 14wt% CBs 4wt% | 3.32 | 830.49±102.06 | 0.027±0.082 |

**Fig. 15.** SEM images of PAN/CBs nanofiber membranes under different total polymer concentration of (a) PAN 8 wt% + CBs 4 wt%, (b) PAN 8 wt% + CBs 6 wt%, (c) PAN 8 wt% + CBs 8 wt%, (d) PAN 8 wt% + CBs 10 wt%, (e) PAN 10 wt% + CBs 2 wt%, (f) PAN 10 wt% + CBs 4 wt%, (g) PAN 10 wt% + CBs 6 wt%, (h) PAN 10 wt% + CBs 8 wt%, (i) PAN 14 wt% + CBs 2 wt%, and (j) PAN 14 wt% + CBs 4 wt%. Overlaid on its images are the EDS mapping analysis. The corresponding fiber diameter distributions are shown in Figure (S14, S16, and S18(a, b)), whereas the pore-size distributions are presented in Figure (S15, S17, and S18(c, d)). Inverted Table 4 Specific Surface Area (BET), and evaluation of the measured average diameter and pore size.

Post-filtration SEM images (Fig. 14(d–f)) reveal that membranes fabricated at higher PAN concentrations appear less loaded with captured particles. This is not indicative of poor filtration performance, but rather stems from two interrelated factors: (i) the significantly larger fiber diameter reduces specific surface area, limiting available adsorption sites; and (ii) particles are more likely to be trapped within the deeper layers of the porous network or between fibers, rather than adhering to the surface, rendering them less visible in top-view SEM. Additionally, the smoother surface and lower surface energy of thick fibers may reduce particle adhesion, allowing some particles to rebound rather than deposit. Supporting this interpretation, BET surface area measurements (Inverted Table 3 in Fig. 14) show a marked decline with increasing polymer concentration: 8.97 $m^2/g$ (12 wt%), 5.33 $m^2/g$ (16 wt%), and 2.88 $m^2/g$ (18 wt%). The inverse relationship between fiber diameter and specific surface area directly explains the reduced apparent particle coverage in SEM. Furthermore, EDS elemental mapping of pure PAN membranes (Fig. 14(g–i)) shows an increasing nitrogen (N) signal intensity with higher polymer concentration, consistent with greater PAN mass per unit area. This confirms successful fiber formation across the concentration range and validates the compositional homogeneity of the membranes. Therefore, these results demonstrate that while higher PAN concentrations produce thicker fibers and more open pore structures, beneficial for reducing pressure drop, the associated loss in specific surface area and altered particle deposition behavior must be carefully balanced against filtration efficiency requirements. Optimal membrane design thus requires tuning concentration to achieve a trade-off between pressure drop, structural integrity, and particle capture capability.

Further morphological investigations were conducted on PAN/CBs composite nanofibers to elucidate the impact of cigarette butt-derived material (CBs) incorporation under fixed PAN concentrations. Based on the electrospinnability window established in this study, spanning total polymer concentrations from 8 wt% to 18 wt%, PAN solutions at 8 wt%, 10 wt%, and 14 wt% were selected as base formulations. CBs were then added incrementally to raise the total solids concentration (PAN + CBs) up to 18 wt%, enabling systematic evaluation of high-solids electrospinning feasibility. Upon CBs addition, a progressive increase in fiber diameter was observed across all PAN base concentrations: for PAN 8 wt%+CBs, diameters rose from 274.71 nm to 874.95 nm (Fig. 15(a–d)); for PAN 10 wt%+CBs, from 284.16 nm to 841.62 nm (Fig. 15(e–h)); for PAN 14 wt%+CBs, from 763.68 nm to 830.49 nm (Fig. 15(i–j)), whereas their fiber diameter distribution are showed in Figs. S14, S16, and S17(a, b). This fiber thickening is primarily attributed to the sharp increase in solution viscosity resulting from higher total solids content. Although CBs may enhance electrical conductivity, which typically promotes jet elongation, the dominant effect of elevated viscosity restricts chain mobility and suppresses jet thinning during electrospinning. At high CBs loadings, intensified polymer chain entanglement and particle crowding further reduce jet drawability, outweighing any conductive benefits. Concurrently, average pore size decreased monotonically with increasing total solids concentration across all PAN/CBs systems. This reflects denser fiber packing and reduced inter-fiber spacing, consistent with the formation of thicker, less flexible fibers that settle more compactly on the collector.

Membrane thickness exhibited a non-linear trend with total solids concentration (Fig. 16(a) and Figure S19 (a)): it increased from 8 wt% up to 16 wt% total solids but decreased at 18 wt%. This counterintuitive reduction at the highest concentration arises from two competing effects: (i) enhanced deposition rate due to higher polymer content, and (ii) suppressed fiber whipping instability caused by excessive viscosity, which limits lateral spreading of the jet. The net result is a more compact, densely packed mat with reduced loft, leading to thinner overall membrane thickness despite greater mass loading, a behavior previously observed in highly viscous

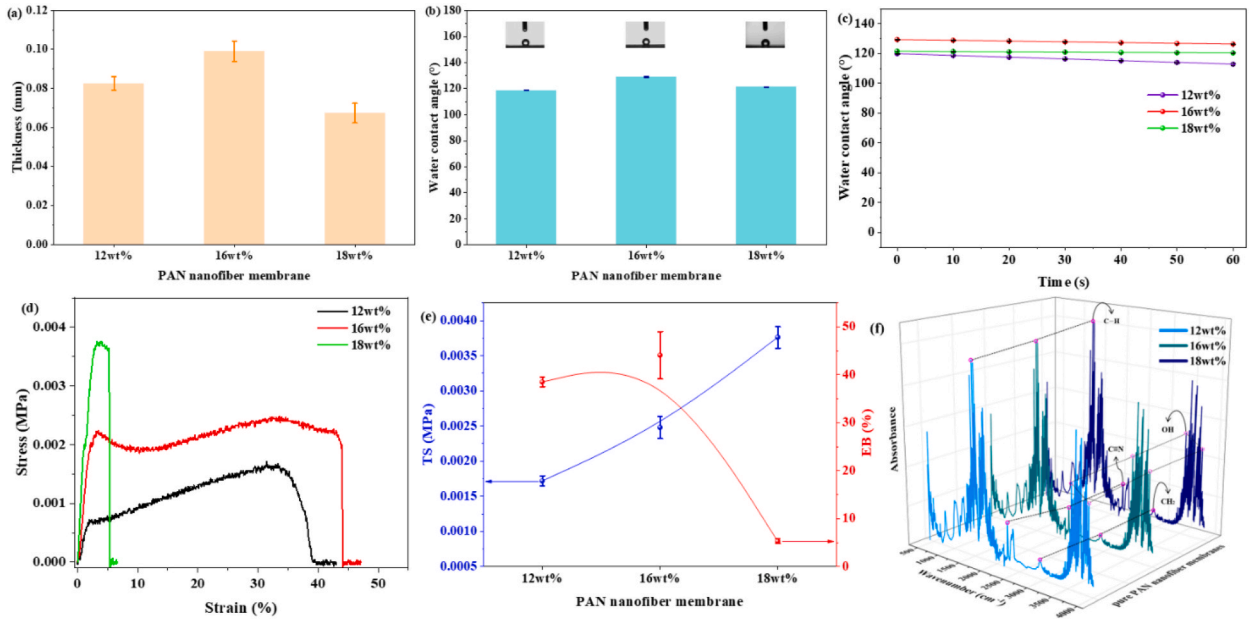



**Fig. 16.** (a) Membrane thickness, (b) water contact angle, (c) water contact angle over 60s, (d) stress-strain curves, (e) tensile stress (TS) and elongation at break (EB), and (f) FT-IR spectra of pure PAN nanofiber membranes at 12 wt%, 16 wt% and 18 wt% total concentration.

electrospinning systems. Notably, smooth, continuous, and bead-free fibers were successfully fabricated even at 18 wt% total solids, confirming the feasibility of high-concentration electrospinning when CBs are used as a co-component. This contrasts sharply with earlier experiments in which total solids concentration was held constant while the PAN:CBs ratio was varied; in those cases, CBs incorporation led to thinner fibers, likely due to improved charge carrying capacity and enhanced jet stretching. This apparent contradiction underscores the context-dependent role of CBs: at fixed total concentration, CBs act as a conductive or rheology-modifying additive that promotes jet elongation, yielding finer fibers. At increasing total solids, CBs primarily contribute to viscosity buildup, which dominates over conductive effects and results in fiber thickening. Inverted Table 4 (in Fig. 14) summarizes the evolution of fiber diameter, pore size, and specific surface area across the three PAN base systems as total solids increase. Consistent with morphological trends, BET surface area decreases significantly with rising CBs content: from 8.17 to 4.34 $m^2/g$ for PAN 8 wt% + CBs, from 8.65 to 4.54 $m^2/g$ for PAN 10 wt% + CBs, and from 4.91 to 3.32 $m^2/g$ for PAN 14 wt% + CBs. This decline directly correlates with increasing fiber diameter, as thicker fibers inherently possess lower surface-to-volume ratios. The reduction in specific surface area aligns with fundamental geometric principles and explains the diminished apparent particle adhesion observed in post-filtration SEM imaging. Based on aforementioned analysis, while CBs enable processing at high total solids concentrations without bead formation, their impact on filtration-relevant properties, such as pore structure, surface area, and fiber morphology, is governed by the overall solution rheology rather than CBs content alone. Therefore, optimizing PAN/CBs membranes for air filtration requires careful balancing of total solids concentration to avoid excessive fiber thickening and surface area loss, particularly when targeting applications demanding high adsorption capacity or fine particle capture.

The surface wettability of the electrospun membranes was evaluated via WCA measurements. As shown in Fig. 16(b), pure PAN membranes fabricated at 12, 16, and 18 wt% exhibit strong hydrophobicity, with WCAs ranging from 118° to 129°. The water droplets remain nearly unchanged over a 60-s observation period (Fig. 16(c)), confirming the stability of the hydrophobic surface and negligible water absorption under ambient conditions. Similarly, PAN/CBs composite membranes also display hydrophobic behavior (Fig. S19(b)). A slight decrease in WCA is observed over time for all CBs-containing samples (Fig. 17(a)), indicating minimal surface interaction or slow uptake of trace moisture. However, the magnitude of this change is small, and all membranes retain WCAs well above 90°, preserving their overall hydrophobic character. This behavior can be attributed to two factors: (i) the inherent hydrophobicity of cellulose acetate derived from cigarette butts, enhanced by residual acetyl groups that reduce surface polarity compared to standard cellulose acetate; and (ii) the dominance of PAN on the fiber surface, as CBs are incorporated at relatively low loadings and do not significantly alter the surface chemistry of the composite fibers.

Concurrent evaluation of mechanical performance reveals a clear dependence on polymer concentration and CBs incorporation. Fig. 16(d) presents the stress–strain curves of pure PAN membranes at 12, 16, and 18 wt%. Tensile strength increases with concentration, 0.0017 MPa (12 wt%), 0.0025 MPa (16 wt%), and 0.0038 MPa (18 wt%), reflecting enhanced fiber cohesion and network integrity at higher solids content. However, elongation at break shows a non-monotonic trend: it rises from 38.5% (12 wt%) to 44.1% (16 wt%), but drops sharply to 5.3% (18 wt%), consistent with the formation of stiffer, more brittle fibers at elevated viscosity. In contrast, the addition of CBs generally leads to a modest reduction in tensile strength but introduces notable changes in ductility. For instance, the PAN 8 wt%+CBs 10 wt% and PAN 10 wt% + CBs 8 wt% membranes exhibit tensile strengths of 0.0045 MPa and 0.0032 MPa, respectively, accompanied by limited elongation (7.5% and 10.1%) (Fig. 17(b)). Other CBs-containing formulations follow similar trends, as summarized in Fig. 17(c). The slight strength enhancement in the PAN 8 wt%+CBs 10 wt% sample, despite its low PAN content, may arise from improved fiber packing or interfacial interactions between PAN and CBs-derived components. Nevertheless, the overall reduction in strain-to-failure across most composites suggests that CBs act as rigid fillers that restrict polymer chain mobility, thereby increasing stiffness at the expense of flexibility. Together, these results indicate that while CBs incorporation preserves the hydrophobic nature of PAN nanofibers and can marginally enhance mechanical strength under specific formulations, it concurrently reduces membrane ductility and specific surface area (as discussed previously). The trade-offs among wettability, mechanical robustness, and filtration-relevant morphology must therefore be carefully balanced when designing sustainable air filters from waste-derived additives like CBs.

Finally, FT-IR analysis further confirms that the characteristic absorption of both pure PAN (Fig. 16(f)) and PAN/CBs (Fig. S20) blend membranes appear at identical wavenumbers, indicating no significant chemical bonding or structural alteration, as discussed

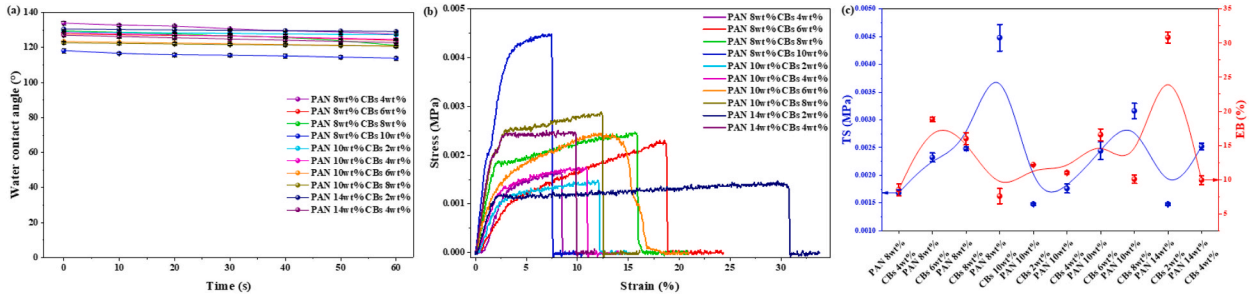



**Fig. 17.** (a) Time-dependent water contact angle over 60 s, (b) representative stress-strain curves, and (c) tensile stress (TS) and elongation at break (EB) of PAN/CBs nanofiber membranes prepared at constant PAN concentrations of 8, 10, and 14 wt% with varying total polymer concentrations. In addition, the corresponding membrane thickness and water contact angle can be found in Fig. S19.

earlier. However, peak sharpness increases with higher total polymer concentration, suggesting enhanced molecular ordering or reduced chain mobility.

### 3.3. Techno-economic performance evaluation of PAN/CBs nanofiber membranes across variable airflow conditions

The spider web charts in Fig. 18 present a comprehensive techno-economic analysis comparing the performance and cost-effectiveness of PAN/CBs hybrid nanofiber membranes against a baseline pure PAN (14 wt%) membrane under three distinct airflow velocities: 3.3 cm/s, 5.3 cm/s, and 7.3 cm/s. The evaluation integrates key filtration metrics, including filtration efficiency (η), pressure drop, maximum dust-holding capacity, quality factor (QF), and material cost, to assess the trade-offs between performance and sustainability across varying operational conditions.

At the lowest airflow velocity of 3.3 cm/s, the PAN/CBs membranes demonstrate significant techno-economic advantages over the pure PAN reference. Notably, blends with higher CBs content exhibit substantially enhanced dust-holding capacities. For instance, the PAN 10 wt% + CBs 6 wt% formulation (Fig. 18(d)) achieves a maximum dust capacity ratio of 2.07, while PAN 8 wt% + CBs 10 wt% (Fig. 18(a)) reaches 1.65, indicating more than double the service life before reaching the critical 450 Pa pressure drop threshold. These improvements are achieved without compromising filtration efficiency, as η ratios remain above 0.998, comparable to the baseline. Moreover, the incorporation of low-cost, waste-derived CBs reduces material costs by approximately 26-27% (cost ratios ≈ 0.73-0.74). While some formulations show elevated initial pressure drops (e.g., PAN 8 wt% + CBs 4 wt%, pressure drop ratio = 3.43), optimal compositions such as PAN 10 wt% + CBs 6 wt% and PAN 10 wt% + CBs 8 wt% strike an excellent balance: moderate pressure increase (ratios of 1.28–1.11), high QF ratios (0.97 and 0.79), and superior dust retention. These results highlight their strong potential for low-velocity applications, such as indoor air purification systems and residential HVAC filters.

At 5.3 cm/s, performance trends shift due to increased aerodynamic stress. The PAN 10 wt% + CBs 6 wt% membrane (Fig. 18(e)) remains the most promising, maintaining a high dust capacity ratio (1.47) and a favorable QF ratio (0.65), albeit reduced from lower velocities. In contrast, PAN 8 wt% + CBs 8 wt% (Fig. 18(b)) shows a more modest improvement in dust capacity (0.83×) and a lower QF ratio (0.34), while PAN 8 wt% + CBs 10 wt% underperforms with only half the dust capacity of the reference. Nevertheless, all tested blends continue to offer 18–27% cost savings, and filtration efficiency remains virtually unchanged (η ratios ≈ 0.999-1.001). Pressure drop ratios increase moderately (1.08-1.69), suggesting that although these membranes remain functional at moderate flow rates, their resistance to clogging and overall filtration quality decline relative to low-velocity operation. This indicates a sensitivity to airflow dynamics, where structural integrity and pore stability become critical factors.

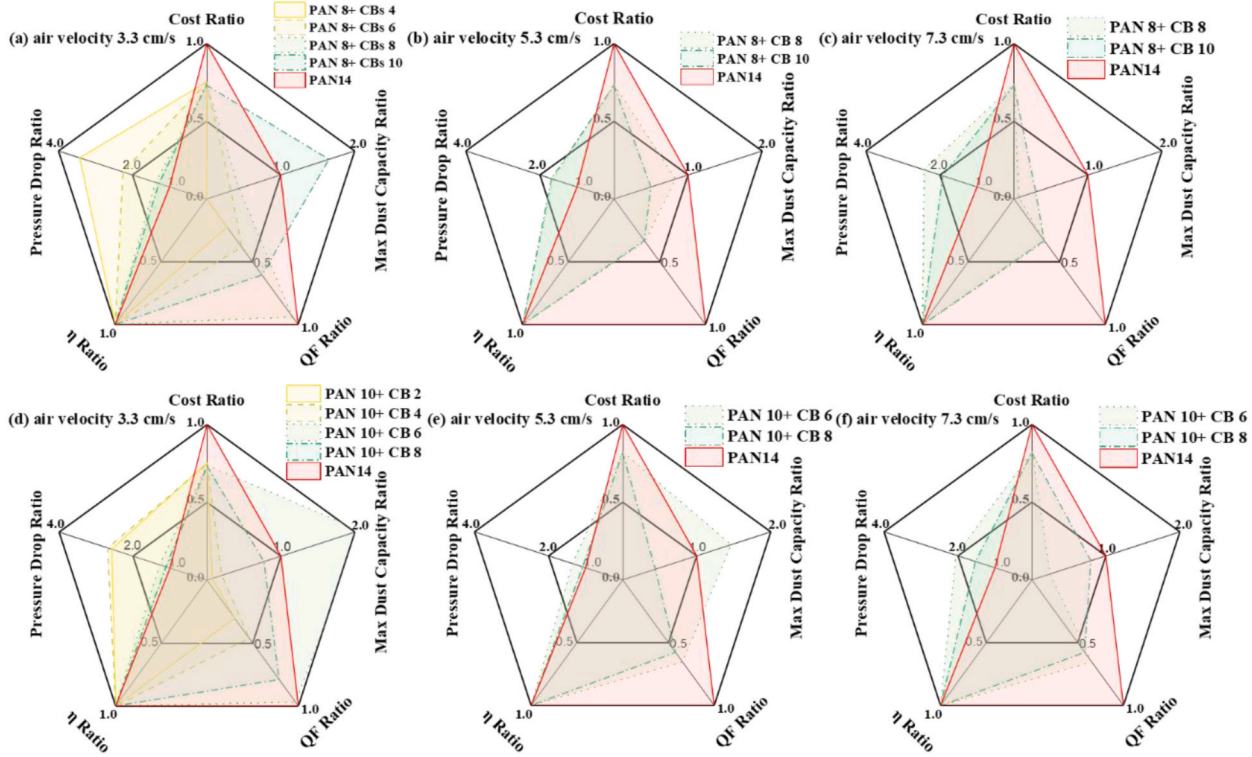



**Fig. 18.** Spider web plots comparing techno-economic performance of PAN/CBs nanofiber membranes versus pure PAN 14 wt% under three airflow velocities of 3.3 (a, d), 5.3 (b, e), and 7.3 (c, f) cm/s. Parameters include filtration efficiency (η), pressure drop, maximum dust capacity, quality factor (QF), and material cost ratio. Each subplot corresponds to a specific blend composition, highlighting trade-offs between performance and sustainability.

At the highest tested velocity of 7.3 cm/s, the dust-holding performance of most PAN/CBs membranes deteriorates significantly. As shown in Fig. 18(c) and (f), the previously top-performing PAN 10 wt% + CBs 6 wt% blend now exhibits only 24.5% of the reference's dust capacity, while PAN 8 wt% + CBs 8 wt% drops to just 6.7%. Interestingly, the PAN 10 wt% + CBs 8 wt% formulation demonstrates relatively better resilience, retaining 80% of the baseline dust capacity, suggesting that higher CBs loading may enhance structural robustness under high-stress conditions. However, its QF ratio remains limited at 0.58. Despite this, all blends maintain excellent filtration efficiency (η ratios >1.001) and substantial cost reductions (cost ratios 0.73-0.82). Notably, pressure drop ratios rise sharply, up to 2.44 for PAN 8 wt% + CBs 8 wt%, indicating rapid fouling and shortened operational lifespan.

Therefore, this techno-economic assessment reveals that the integration of upcycled cigarette butts into PAN-based nanofiber membranes not only reduces production costs but also enhances dust-holding performance under favorable operating conditions. The optimal composition, PAN 10 wt% + CBs 6 wt%, emerges as a particularly promising candidate for sustainable air filtration systems, especially in indoor settings where airflow is typically low to moderate. The results support the viability of waste-derived materials in high-performance filtration applications and emphasize the importance of matching membrane design to specific operational demands.

### 3.4. Stability, recyclability, and comparative assessment of CBs-derived nanofiber membranes

The sustained performance over time and across varying flow rates indicates that the nanofiber membrane is highly resistant to rapid clogging and structural degradation, which are typical limitations in ultrafine particle filtration. Accordingly, the long-term $PM_{1.0}$ purification efficiencies of the PAN (10 wt%) + CB (6 wt%) nanofiber membranes were evaluated, as depicted in Fig. 19(a). After 130 min of continuous operation, the membrane maintained a remarkable $PM_{1.0}$ removal efficiency of 99.98% across all tested air flow velocities (3.3, 5.3, and 7.3 cm/s), demonstrating both its robust particle capture capability and reliable operational stability. Moreover, to evaluate their performance under challenging environmental conditions, the PAN (10 wt%) + CB (6 wt%) nanofiber membranes were exposed to extreme temperatures (−50°C and 85°C, 2 h each), prior to $PM_{1.0}$ filtration testing at 75% relative humidity across the aforementioned air flow velocities. After such environmental conditioning, the $PM_{1.0}$ removal efficiency remained exceptionally high, ranging from 99.98% to 99.995%, accompanied by a modest increase in pressure drop (approximately 5-12%), as shown in Fig. 19(b). This slight increase in pressure drop is likely attributable to minor fiber relaxation at elevated temperatures and transient moisture adsorption on the fiber surface. These findings can be explained by the thermomechanical behavior of the

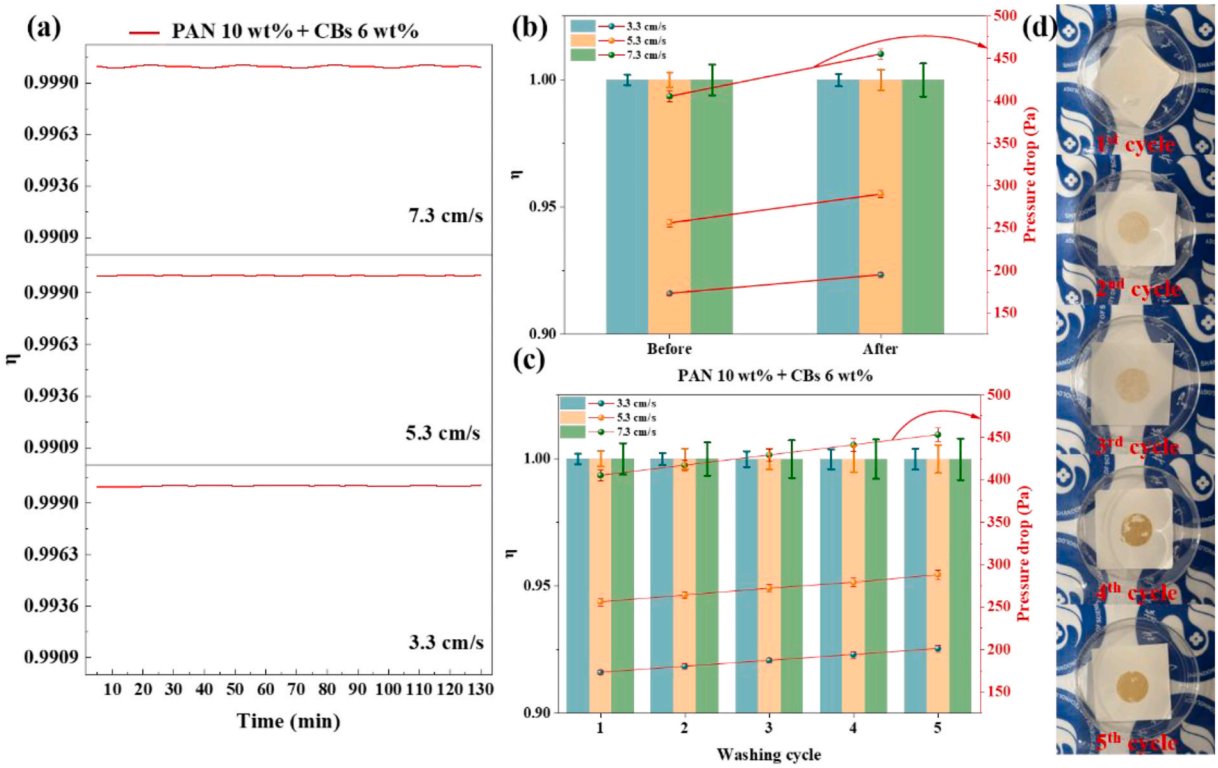



**Fig. 19.** $PM_{1.0}$ purification performance of the PAN (10 wt%) + CB (6 wt%) nanofiber membrane. (a) Long-term filtration efficiency during a 130-min continuous test. (b) $PM_{1.0}$ removal efficiency before and after exposure to extreme temperatures (−50°C and 85°C for 2 h each) and subsequent testing at 75% relative humidity. (c) Filtration efficiency after five washing cycles, and (d) photographs of the membrane obtained after each washing cycle.

membrane: when heated to 85°C, the PAN polymer softens slightly, allowing the ultrafine fibers to pack more closely, which may induce minor intermolecular bonding and a marginal reduction in pore size. Conversely, when cooled to −50°C, the fibers temporarily stiffen but regain their normal flexibility upon returning to room temperature without incurring permanent structural damage. Furthermore, under high humidity, surface electrostatic charges, which assist in particle capture, may partially dissipate, and a thin adsorbed water layer can form on the membrane surface. Nevertheless, given that the filtration process is primarily driven by mechanical mechanisms (i.e., diffusion and interception within the dense nanofiber network), the overall performance remains robust even when electrostatic contributions diminish.

The reusability of the PAN (10 wt%) + CB (6 wt%) nanofiber membrane was evaluated through five consecutive filtration-regeneration cycles. Following each cycle, the membrane was regenerated using a mixture of deionized water and anhydrous ethanol. Subsequently, both the filtration efficiency and pressure drop were measured under standard ambient conditions at airflow velocities of 3.3, 5.3, and 7.3 cm/s, as illustrated in Fig. 19(c)–(d). The membrane demonstrated exceptional durability throughout repeated use, maintaining a $PM_{1.0}$ filtration efficiency of approximately 99.97% across all tested airflow velocities even after five complete cycles. Concurrently, the pressure drop exhibited a modest increase of approximately 10-15%. These minor variations are primarily attributed to slight fiber rearrangement, partial dissipation of surface electrostatic charges during wet cleaning, and the accumulation of residual particulates within the pore structure, collectively leading to a progressive reduction in air permeability. Nevertheless, given that particle capture is predominantly governed by mechanical trapping within the dense nanofiber network, the overall filtration performance remains highly stable.

To benchmark the air filtration performance of the PAN (10 wt%) + CB (6 wt%) nanofiber membrane against other waste-derived alternatives, the optimal filtration efficiency, pressure drop, and quality factor of various electrospun nanofibrous membranes are summarized in Table 1. Evaluated across these key parameters, the overall performance of the PAN (10 wt%) + CB (6 wt%) membrane is highly competitive with state-of-the-art counterparts. Notably, the proposed membrane achieves an approximately 27% reduction in production costs. Furthermore, while previous studies primarily focus on the initial utilization of waste materials, they often lack a comprehensive circular economy solution for the continuous recovery and valorization of waste-derived components during the membrane fabrication process.

**Table 1**
Air filtration performances of waste-derived electrospun nanofibrous membranes.

| Materials | Airflow velocity (cm/s) | Particulate matter | Filtration efficiency | Pressure drop (Pa) | QF ($Pa^{-1}$) | Cost reduction (%) | Refs. |
|---|---|---|---|---|---|---|---|
| PAN/Fly ash (lignate) | - | TVOCs ($mg/m^3$) | 59.0% | - | - | - | Zehra et al. (2025) |
| PLA/lignin (LIG) particles | 32 L/min | ≤0.3 (incense smoke) | 99.6% | 105 | 0.073 | - | Scaffaro and Citarrella (2024) |
| EPS waste | 5.3 | ≤2.5 (incense smoke) | 99.99% | 145 | 0.15 | - | Rajak et al. (2020) |
| Recycled PET_50Ag | 15.83 | ≤1.0 (KCl aerosol) | 99.97% | - | - | - | Kijeńska-Gawrońska et al. (2025) |
| Recycled PET/BAC | 85 L/min | ≤0.025-0.15 (NaCl aerosol) | 98.8% | 550-623 | 0.0071 | - | Russell and Jana (2025) |
| Recycled PET | 4.8 | ≤0.3 (bioaerosol) | 98.4% | 212 | 0.0195 | - | Bonfim et al. (2021) |
| Waste plastic bottles (PET) | 28-77 | ≤1.0 (aerosol particles generator) | 99.7% | 11.56 Pa/$cm^2$ | 0.03 | - | Kakoria et al. (2021) |
| Recycled TPU&PLA | - | ≤2.5 (burning tissue paper smoke) | 99.0% | 140 | 0.0329 | - | Chambers et al. (2024) |
| HIPS waste | 5.3 | ≤2.5 (incense smoke) | 98.75% | - | - | - | Zulfi et al. (2019a) |
| Recycled PS and PET | 4.8 | ≤85 nm particles sizes from (aerosol particles generator) | 94.49% | 58.5 | 0.0448 | - | Lima et al. (2026) |
| Recycled CBs | 10-96 | ≤2.5 (aerosol particles generator) | 99.99% $PM_{2.5}$ 99.7% $PM_{0.1}$ | 50.4 | 0.06 | - | Kakoria and Sinha-Ray (2022) |
| Waste-acrylonitrile butadiene styrene (ABS) | 5.3 | ≤2.5 (NaCl aerosol) | 96.17% | 86.6 | 0.0376 | - | Munir et al. (2022) |
| Textile waste rPA/MAG | 5 | ≤0.1-2.5 (aerosol particles generator) | 96% | 189 | 0.024 | - | Opálková Šišková et al. (2022) |
| ABS waste | 5.3 | ≤2.5 (aerosol particles generator) | 95.16% | 118.7 | 0.0255 | - | Zulfi et al. (2019b) |
| Food waste BPE, PVA | 0.01 | ≤2.5 (NaCl aerosol) | 98.3% | 156 | 0.026 | - | Agrawal and Nawaz (2025) |
| PAN/CBs | 5.3 | ≤1.0-10 (incense smoke) | 99.99% $PM_{1.0}$ | 256 | 0.03759 | 18-27% | This work |
| PAN/CBs | 3.3 | ≤1.0-10 (incense smoke) | 99.99% $PM_{1.0}$ | 173 | 0.05576 | 26-27% | This work |

### 3.5. A sustainable circular approach: transforming discarded CBs into high-performance nanofiber membranes for air and water filtration

The development of hybrid polyacrylonitrile/cellulose acetate from cigarette butts (PAN/CBs) nanofiber membranes represents a sustainable and circular strategy to address the global challenge of cigarette butt waste, one of the most prevalent forms of litter worldwide. As illustrated in Fig. 20, this study presents a closed-loop process that converts discarded CBs, commonly found in public spaces, beaches, and landfills, into high-performance filtration materials, thereby transforming environmental pollutants into valuable resources. The process begins with the collection of used CBs from urban and natural environments, preventing their accumulation in ecosystems and reducing pollution. Collected CBs undergo chemical pretreatment to recover CA, the primary component of cigarette filters. This involves sorting, washing with absolute ethanol to remove tar, nicotine, and other toxic residues, followed by rinsing and drying. Critically, the ethanol used in this step is recovered via distillation and reused, minimizing solvent waste and supporting green processing principles. The recovered CA is then blended with PAN as a renewable, low-cost additive and electrospun into nanofiber membranes. This approach not only reduces reliance on virgin polymers but also enables the upcycling of waste into functional materials through scalable and sustainable fabrication. The resulting PAN/CBs nanofiber membranes exhibit excellent structural and surface properties suitable for dual applications in air and liquid filtration. In air filtration, the as-spun membranes demonstrate strong hydrophobicity (water contact angle $>120^{\circ}$), high porosity, and a fine fibrous network, enabling efficient capture of airborne particulate matter, including $PM_{1.0}$, $PM_{2.5}$, and $PM_{10}$, with minimal pressure drop. These characteristics make them ideal candidates for use in indoor air purifiers, HVAC systems, and personal protective face masks, offering effective protection against airborne pollutants. For liquid filtration, zeta potential measurements (Figure S21 and Table S6) reveal that the membranes possess a predominantly negative surface charge across a wide pH range (pH 5-9), particularly under neutral and alkaline conditions. This intrinsic negative charge enhances electrostatic repulsion against negatively charged contaminants such as bacteria, organic molecules, and colloidal particles commonly present in wastewater and natural water sources. Consequently, the membranes exhibit improved fouling resistance and stable filtration performance, making them highly suitable for microfiltration and ultrafiltration applications. Furthermore, the retained negative charge facilitates compatibility with thin-film composite (TFC) membrane fabrication (Gohain et al., 2024; Luthfiana et al., 2025; Mahto et al., 2022). A simple NaOH post-treatment renders the support layer hydrophilic, enabling uniform deposition of a polyamide selective layer, a critical step in producing high-flux, durable TFC membranes. In membrane distillation (MD) for desalination, the hydrophobic nature of the PAN/CBs membranes (WCA $>120^{\circ}$) ensures efficient separation of clean water vapor from saline or contaminated feed solutions. Their robust water-repellent surface prevents wetting, even under harsh operating conditions, thereby enhancing long-term stability and performance in saline environments. Finally, at end-of-life, the membranes can be collected and recycled either through chemical reclamation or repurposing, completing the circular lifecycle. This closed-loop system, from waste collection to membrane fabrication, application, and recycling, demonstrates a full circular economy model: waste CBs → green processing → high-value membranes → dual air and water filtration → reuse/recycling. By converting a ubiquitous pollutant into multifunctional filtration materials, this work advances the integration of waste management and environmental technology. It offers a practical, scalable, and environmentally responsible solution that simultaneously supports cleaner air, sustainable water treatment, and reduced plastic pollution. The proposed strategy exemplifies how circular innovation can drive both environmental stewardship and technological advancement, aligning closely with the goals of resource conservation and sustainable development.

## 4. Concluding remarks and future prospect

This study demonstrates that discarded cigarette butts can be upcycled into high-performance hybrid nanofiber membranes for air filtration, effectively transforming a ubiquitous plastic waste stream into a valuable raw material. The integration of cigarette butt-derived cellulose acetate with polyacrylonitrile *via* blend electrospinning establishes a scalable pathway that simultaneously enhances filtration performance and reduces raw material costs. The optimal formulation (10 wt% PAN and 6 wt% CBs) achieves near-complete removal of fine particulate matter while maintaining a low pressure drop, validating both the technical and economic feasibility of integrating waste-derived polymers. Importantly, this performance enhancement is accompanied by a 27% reduction in raw material costs compared to virgin PAN, highlighting the strong economic incentives for utilizing waste-derived polymers. Furthermore, integrating comprehensive filtration testing with a techno-economic analysis of CB-based air filters provides critical evidence supporting their industrial scalability. Therefore, the present findings provide a compelling foundation for scaling CB valorization as a practical and sustainable solution within modern materials management frameworks.

However, the practical implementation of this approach requires overcoming several real-world challenges. First, the nanofiber membrane filters must be thoroughly evaluated to ensure they maintain high performance over extended periods under fluctuating humidity, temperature, and continuous exposure to complex air pollutants. Second, establishing consistent quality control for CB collection, developing scalable production methods, and ensuring the safe disposal or recycling of spent filters are essential to prevent secondary microplastic pollution and comply with environmental regulations. Finally, although heat-pressing enhances the mechanical robustness of the membranes, this post-treatment step requires further optimization for large-scale manufacturing to ensure structural stability under continuous airflow. Therefore, future research should prioritize full-scale life cycle assessments to evaluate the true environmental and economic impacts of these nanofiber membrane filters from production to end-of-life. Pilot-scale manufacturing trials are necessary to increase production throughput, ensure consistent fiber quality, and optimize post-treatment parameters under continuous operation. Additionally, exploring surface functionalization or catalytic doping could further expand the utility of these membranes to include the capture of harmful volatile organic compounds or antimicrobial filtration. Concurrently, collaboration with policymakers is essential to establish performance standards, support reliable recycling supply chains, and integrate

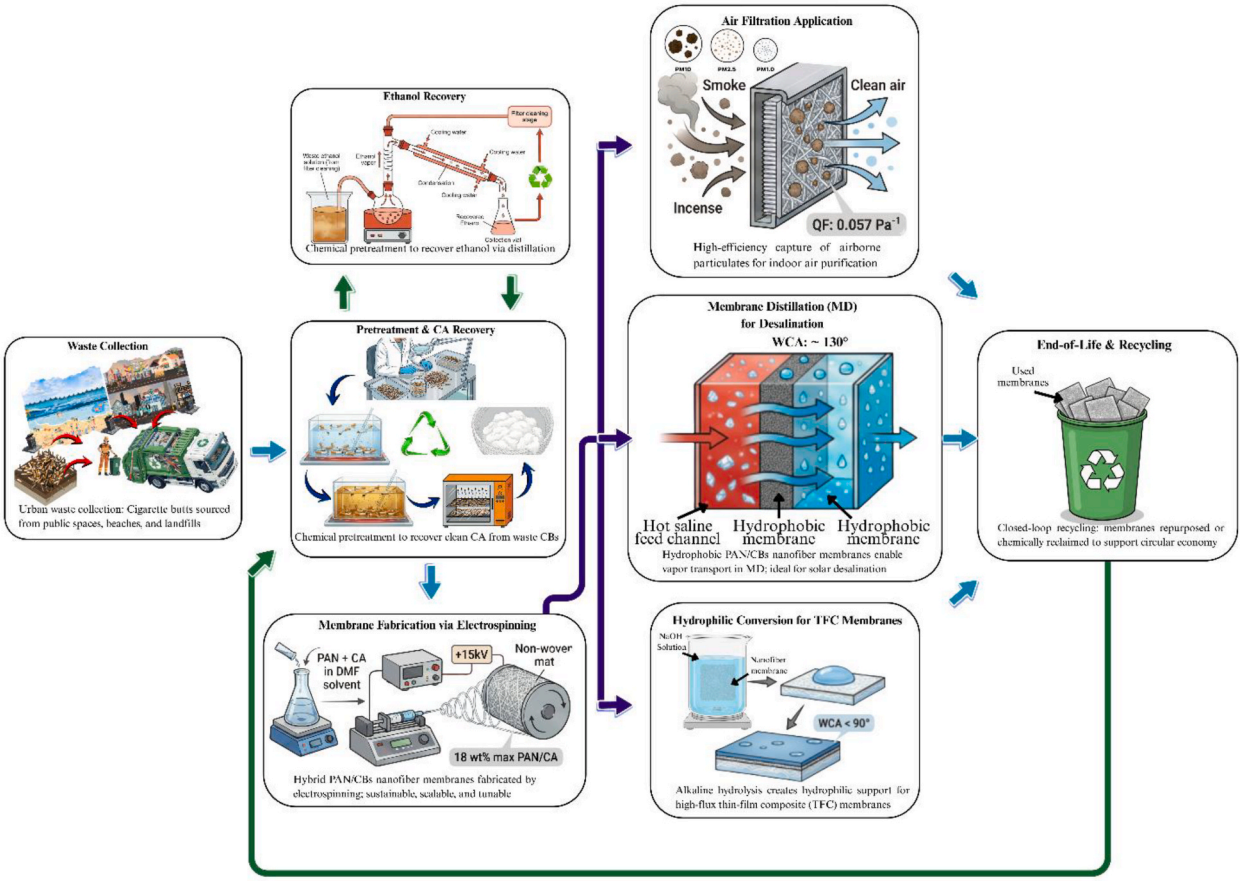



**Fig. 20.** Schematic illustration of the circular lifecycle of PAN/CBs hybrid nanofiber membranes: from collection of discarded CBs in urban and natural environments, through green pretreatment and ethanol recovery, to electrospinning into multifunctional membranes for dual air and liquid filtration applications, and finally to end-of-life recycling, demonstrating a closed-loop, waste-to-resource strategy that integrates environmental remediation with sustainable materials engineering.

cigarette butt collection into extended producer responsibility programs.

## CRediT authorship contribution statement

**Binhasan Akid:** Data curation, Investigation, Methodology, Writing – original draft. **Bo Zheng:** Supervision, Validation, Writing – review & editing. **Changwei Wang:** Resources, Supervision. **Jian Wang:** Resources, Supervision. **Qun Wang:** Conceptualization, Data curation, Methodology, Supervision, Writing – review & editing. **Jun Gao:** Resources, Supervision. **Dongmei Xu:** Resources. **Jianlu Liu:** Resources, Supervision, Validation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scp.2026.102484.

## Data availability

Data will be made available on request.

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